

Features

- 8x Controlled 24V outputs by high-side drivers
- 16x Analog inputs for 3.3v, 5v and 24v amplitude signals with filtering
- CAN communication
- 3x I2C connections
- 2x PWM signal outputs
- 3.3v, 5v and 24v as simple output power supply

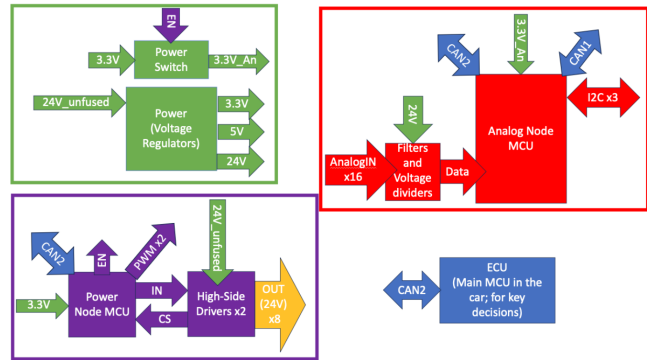


Figure 1: Nodes v2.1 Diagram

Applications

- Receiving and analysing data from a wide variety of sensors
- Controlling and monitoring various functions of the car
- Generation of PWM signal for fans and pumps

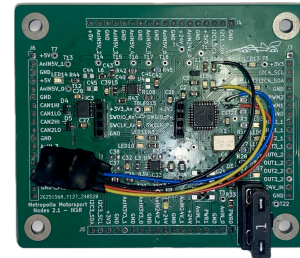


Figure 2: Nodes v2.1 Board Front Side

General Description

In the ever-evolving world of motorsport technology, Nodes are one of the key initiatives of the Metropolia Motorsport team. Aimed at improving the performance and efficiency of the next generation race car, this board focuses on developing a new design for the sensor system. Nodes act as the central nervous system, seamlessly connecting, processing signals, and controlling the various sensors critical to the vehicle's optimal functioning.

The board itself has two main parts Analog Node and Power Node. Analog Node reads analog sensors and sends their value on CANbus, and the Power Node acts as programmable fuses and CANbus controlled switches, as well as having digital and Pulse Width Modulation (PWM) input and output capabilities.

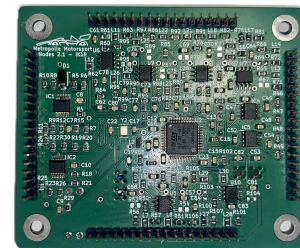


Figure 3: Nodes v2.1 Board Back Side

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Introduction

This datasheet covers all available information on the new Nodes design such as description, overview, electrical characteristics, pin assignment and definition, software. The document itself can be divided into two main parts, which consist of data about the board itself at the electronic level, as well as data regarding the software part.

The new version of Nodes was developed for Metropolia Motorsport and is under constant improvement and updating, so this datasheet should be constantly updated in accordance with the changes, so be careful when choosing the right version of the datasheet to avoid unexpected errors and problems.

This document mainly covers the stable version of Nodes v2.1, which at the time of writing the document is actively used, but the datasheet also contains references and clarifications of the new version 2.2, which was under development and was supposed to replace 2.1.

If the data written in this datasheet is insufficient and forces to explore additional sources of information, this means that it is necessary to supplement the documentation with the missing data, so report this situation to the person responsible for updating this documentation.

Overview

Nodes are inherently an incredible combination of software and a variety of pre-existing components on the market, such as voltage regulators, microchips, operational amplifiers, high-side drivers and others, so it is very important to understand at a basic level what is happening in each part of the board in order to avoid various problems.

Power Supply

In order for Nodes to work reliably, high-quality and stable power is necessary not only for all components on the board, but also for additional connected equipment to the board, for example, sensors. To achieve this goal, a variety of solutions were implemented in Nodes. In both versions (2.1 and 2.2), the 24 volt input voltage is stabilized using diode $D2$, zener diode $D3$ and capacitor $C24$ as shown in Figure 4. As a result of the voltage passing through diode $D2$, a voltage drop occurs by the value V_f , which is presented in Table 7. The board is also protected from high current consumption with popular blade fuse 1A.

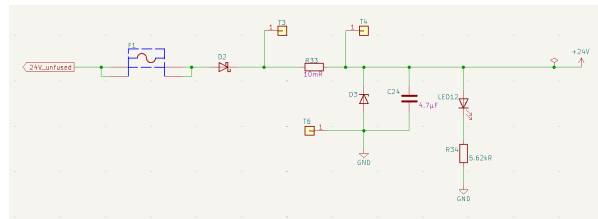


Figure 4: Voltage stabilization for 24V

Resistor $R33$ is used to measure the current consumption of the entire board using two test points $T3$ and $T4$ directly on the board without disconnecting anything. To do this, take a multimeter, put it in voltage measurement mode, and connect it to these test points with two wires, thereby you can get the voltage value through resistor $R33$, in order to convert this into a current value, divide this value by the resistance of the resistor in accordance with the law Ohm, namely, perform the calculation of the Formula 1.

$$I_c = V_R / (10 * 10^{-3}) \quad (1)$$

Measuring current consumption for Analog Node through resistor $R35$ works in the same way; more information about this in Section Power Node.

Voltage Regulators v2.1

After stabilization in version 2.1, 24V turns into 5V (Figure 5) and 3.3V (Figure 6) using switching regulators. 5V powers the components for CAN and is used for output power, and 3.3V powers almost all components on the board and also provides output voltage. This version uses the LT3970[4] as the main voltage regulator, which has been used in many Metropolia Motorsport projects for a long time.

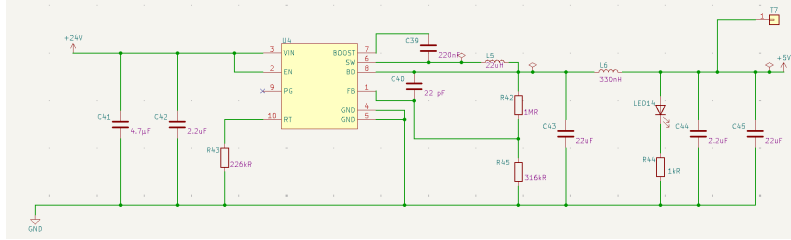


Figure 5: Voltage regulator[4] adj. schematic v2.1 for 5V

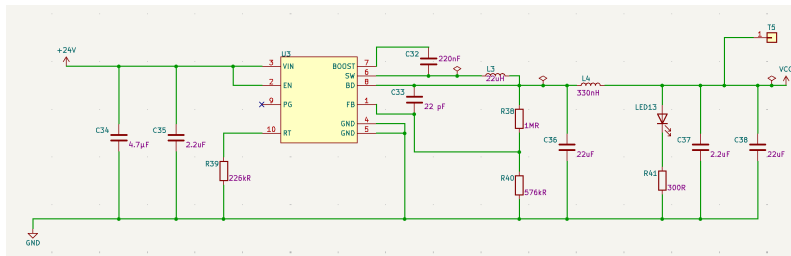


Figure 6: Voltage regulator[4] adj. schematic v2.1 for 3.3V

Some components that are located near the voltage regulators are optional and are not dictated by the datasheet, namely for the 5V voltage regulator: C41, C44, C45 and L6 (L6 can be replaced with a jumper); for a 3.3V voltage regulator: C34, C37, C38 and L4 (L4 can be replaced with a jumper).

Also, the output voltage can be recalculated using Formula 2, in which the values of R_h and R_l could be replaced by ($R42$ and $R45$) or ($R38$ or $R40$) respectively.

$$V_{out} = 1.2 * (1 + R_h/R_l) \quad (2)$$

Based on practical experience, it is recommended to use non-adjustable power components to achieve maximum stability of the result. To use the non-adjustable version of the voltage regulator, some components must be removed.

In the case of 5V, remove C40 and R45, but also replace R42 with a zero resistor.

In the case of 3.3V, remove C33 and R40, but also replace R38 with a zero resistor, so that the resulting connection looks like in the Figure 7.

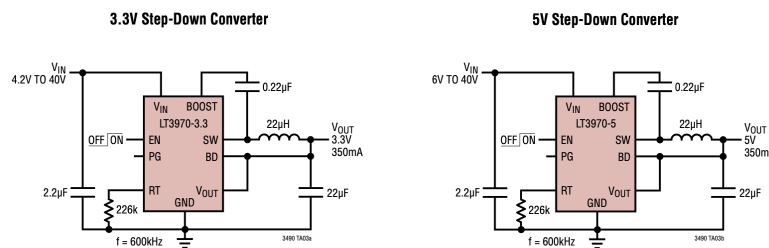


Figure 7: Non-adjustable LT3970 schematics[4]

Over a long period of use, a large number of complaints were collected about this component, such as soldering, reliability, number of additional components, price, etc., so it was decided to find another power solution.

Voltage Regulators v2.2

In version 2.2, the LMR50410-Q1[6] is used as a voltage regulator, which at the time of writing the datasheet seemed the most suitable, given the complaints of the previous version. In this version, there is only one switching regulator on the board, which converts 24V to 5V as shown in Figure 8.

The circuit below on the Figure 8 is for an adjustable version (LMR50410YFQDBVRQ1). And the output voltage can be recalculated using Formula 3.

$$V_{out} = 1 + R_{48}/R_{49} \quad (3)$$

It is recommended to use a fixed voltage regulator (LMR50410Y5FQDBVRQ1).

In order for the circuit to be suitable for a fixed regulator, it is necessary to connect the FB pin to the SW pin directly, so for this remove R_{49} , and solder a jumper in place of R_{48} .

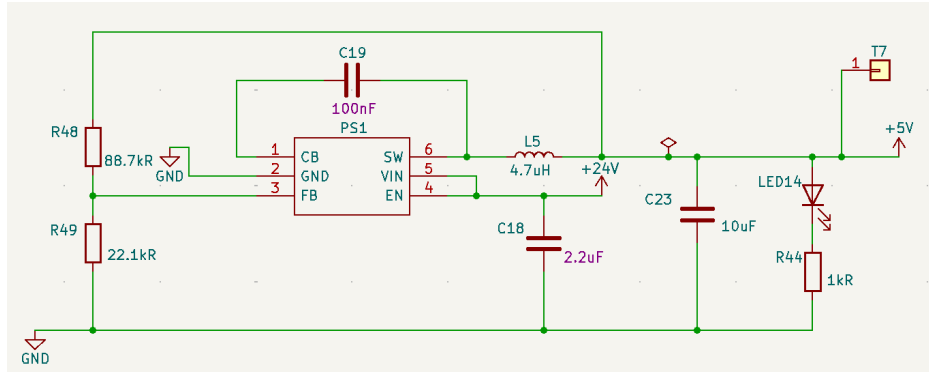


Figure 8: Voltage regulator[6] adj. schematic v2.2 for 5V

The 5V is then converted to 3.3V using the ST730[5] linear regulator to provide even greater 3.3V voltage stability especially for Analog part components (Figure 9).

The circuit below on the Figure 9 is for an adjustable version (ST730MR). If an adjustable version of the component is used, in this case it is recommended to use the Formula 2 to calculate output voltage, where R_h is R_{42} and R_l is R_{43} .

It is recommended to use a fixed voltage regulator (ST730M33R).

In order for the circuit to be suitable for a fixed regulator, remove R_{42} and R_{43} .

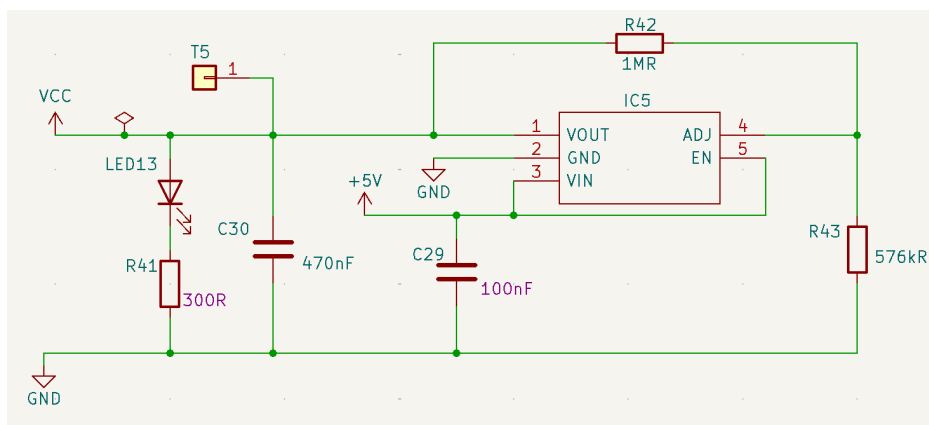


Figure 9: Voltage regulator[5] adj. schematic v2.2 for 3.3V

It is worth noting that there are two such LDOs on the board and one of them is controlled using the $U1$ chip on the Power Node part to control power for the Analog Node. More data on controlled power supply for Analog Node is presented in the next section.

Controllable Power for Analog Node

To be sure of the operation of each part of the Nodes, the control over them is needed, and for this purpose the controllable power supply for Analog Nodes was implemented. In the case of version 2.1, this decision was also caused by the fact that the chip for the Analog Node, unlike the Power Node, was unstable when power was directly connected to it, so the solution with the switch, as shown in the Figure 10, could be described as forced.

But a little later, as a result of conducting numerous tests, it was discovered that the solution with the switch is not a solution to a specific problem, but a treatment for the symptoms caused by it. More information could be found in the Section Power Node.

In the oldest version of 2.1, the power switch TPS22995HQDDCRQ1[7] is not present on the board layout itself, so a separate board was developed, which was soldered into the necessary places (Red Wire - 3.3V; Black Wire - GND; Yellow Wire - Control Signal from Power Node or on the board LED 16; Blue Wire - 3.3V_An), as shown in the Figure 2.

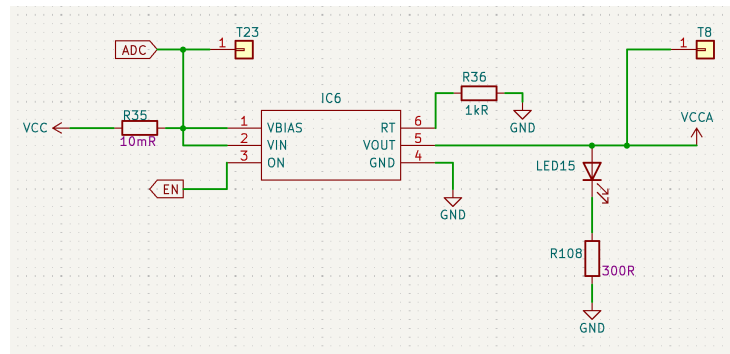


Figure 10: Power Switch[7] for Analog Node power schematic v2.1

Therefore, in the new version 2.2, it was decided to rework the power system, as well as choose a new chip for the Analog Node from the same STM32G4 series as the Power Node, which showed its best side during numerous tests and operations. So the power supply for the Analog Node in this version is not controlled by a separate switch, but by a separate linear voltage regulator[5], which has EN (enable) pin (Figure 11).

The circuit generally follows the pattern shown in Figure 9, with some exceptions. Therefore, here it is also possible to use non-adjustable version with small changes.

The circuit below on the Figure 11 is for an adjustable version (ST730MR). If an adjustable version of the component is used, in this case it is recommended to use the Formula 2 to calculate output voltage, where R_h is $R39$ and R_l is $R40$.

It is recommended to use a fixed voltage regulator (ST730M33R).

In order for the circuit to be suitable for a fixed regulator, remove $R39$ and $R40$.

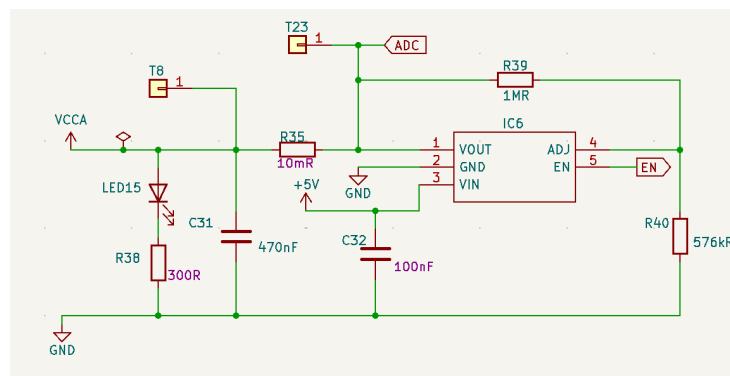


Figure 11: Voltage regulator[5] adj. for Analog Node part schematic v2.2 for 3.3V

Power Node

One of the key parts of the Nodes is the Power Node, which is capable of controlling 24V power for a variety of parts of the car and monitoring their condition with CAN, as well as generating a PWM signal, which is necessary for pumps and fans. The center of it all is the STM32G431KBT6[8] (*U1*) chip. This chip is quite reliable and showed its best performance after numerous tests.

In the Section Controllable Power for Analog Node Power Node actually controls using the EN pin, and also measures the current consumption on this part of the board using pins PA2 and PA3.

Versions 2.1 and 2.2 differ minimally in this section, but the main difference is in how current consumption is measured on the Analog Node. In version 2.1, resistor $R35$, with which the current was measured, was located in front of the switch (Figure 10), and in version 2.2, resistor $R35$ is located after the voltage regulator (Figure 11), which is dictated mainly by nuances associated with the design stage, as well as measurement points, because which seemed much more important to obtain consumption data without the voltage regulator itself. And how to measure current consumption through a resistor is mentioned in the beginning of Section Power Supply.

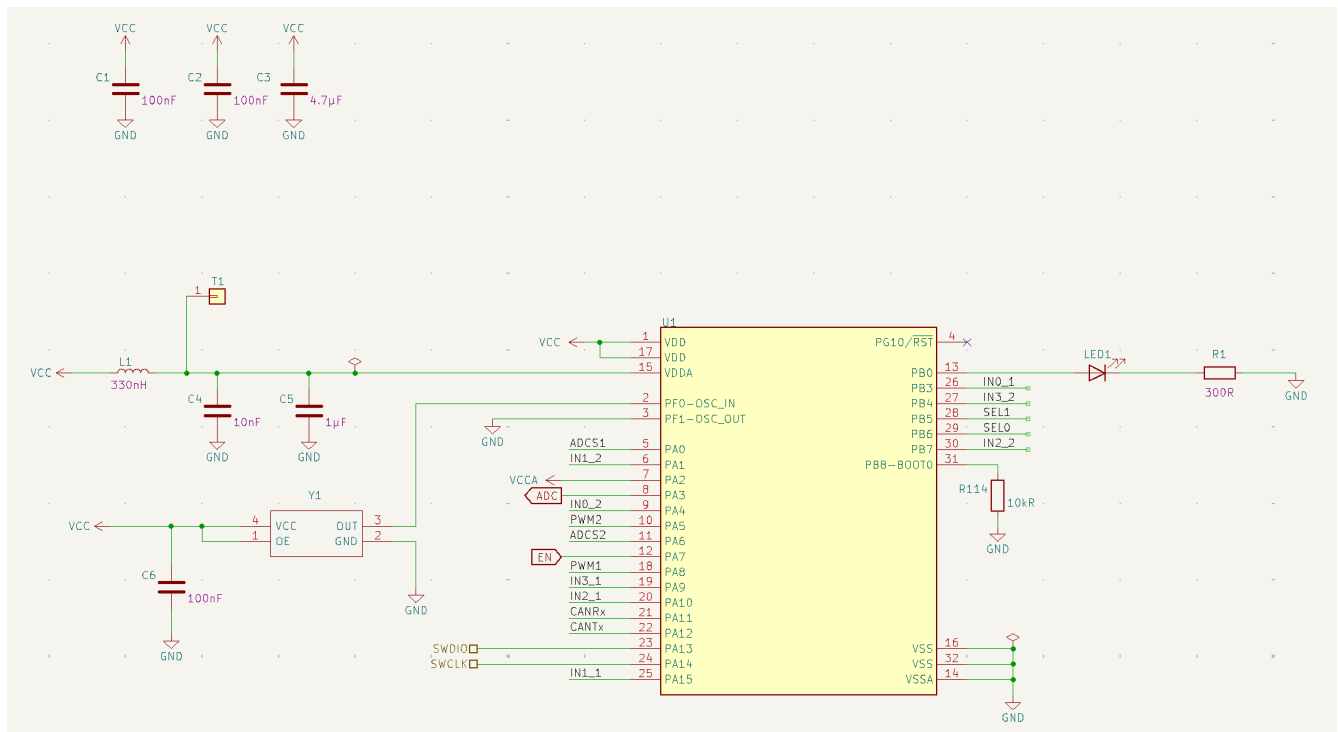


Figure 12: Power Node schematic

For greater stability of the chip, an external oscillator (*Y1*) is used. Power Node version 2.1 uses an oscillator ECS-7050MV-080-BN-TR[9], and version 2.2 uses the same one as the Analog Node SG-8018CE.8.0000M-TJHPA3[10]. Table 1 specifies the frequency range for the external oscillator, but a typical value is 8 MHz. If there is no oscillator with a typical frequency, it is possible to install another one in accordance with the frequency range and footprint sizes (For more information on footprint sizes, see the oscillator datasheet). When installing an oscillator with a different frequency, it is necessary to change the parameters in the code that relate to this, so it is worth notifying the person responsible for editing the code about the change in the oscillator frequency.

There is also a controllable LED (*LED1*) that can be used to monitor the operation of the chip and certain functions. More information about LEDs can be found in Section LED Colour Description.

There should be capacitors near the chip, which should be distributed as evenly as possible around the chip and as close to the pins as possible, all this is done in accordance with the datasheet of the chip itself. In general, this rule applies to all parts of the Nodes, namely, it is necessary to place components from common parts as close as possible to each other, so it is worth paying special attention to the location of particularly critical components on the board, since the stability of the entire board depends on this. The Power Node and Analog Node have a V_{DDA}

pin, which is used by the ADC as a reference, so the analog signal can be converted to digital. A small filter system consisting of an inductor and capacitors is connected to this pin.

It is also worth mentioning that there was a problem with unstable startup of the chip. The failure was previously solved by a banal restart of the power supply of the entire board, which was obviously not the best way. As it turned out later, the unstable operation of the chips during their startup was most likely caused not by unstable power supply, although this may have aggravated the situation, but by the fact that the BOOT pin of the chip remained floating, which, as it turned out, is not acceptable. Therefore, it is strongly recommended to solder a 10k pull-down resistor to each BOOT pin of the STM32 chip, if they are not used in some other way. In version 2.2.3 and newer, there is already a resistor *R114* connected to the BOOT pin, so there is no longer a need to solder large resistors from the outside to solve this problem.

The chip in versions 2.1 and 2.2 is programmed using a JTAG, which consists of four pins, namely +3V3, SWDIO, SWCLK and GND, as shown in the Figure 13. For programming, a special ST-LINK device is used, which is shown in the Figure 14, and STM32CubeIDE software. Connect ST-LINK carefully, because it is possible to make a mistake when connecting and then the power will be connected to ground, which can lead to inevitable consequences.

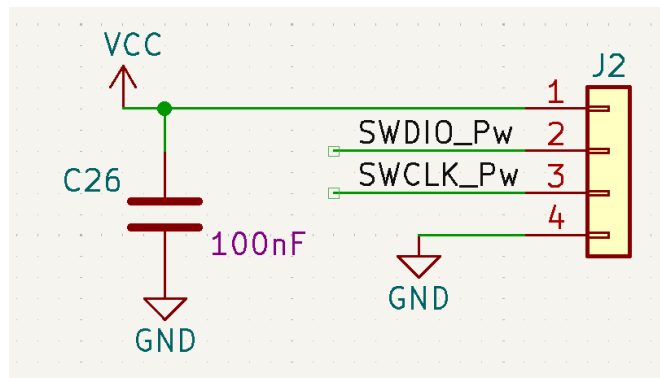


Figure 13: JTAG connector of the Power Node



Figure 14: ST-LINK

Table 1: Power Node (and Analog Node v2.2) Chip[8] Characteristics

Parameter	Symbol	Min.	Typ.	Max.	Unit	Conditions
Supply Voltage	V_{DD}	1.71	3.3	3.6	V	
External Oscillator Frequency	f_{OSC_IN}	4	8	48	MHz	
Current Consumption	$I_{c(MCU)}$	?	?	?	mA	
Voltage Input Low I/O	V_{IL}	-	-	$0.3 * V_{DD}$	V	$1.62V < V_{DD} < 3.6V$
Voltage Input High I/O ¹	V_{IH}	$0.7 * V_{DD}$	-	-	V	
Voltage Output Low I/O ²	V_{OL}	-	-	1.3	V	$ I_{IO} = 20mA$
Voltage Output High I/O ²	V_{OH}	$V_{DD} - 1.3$	-	-	V	$V_{DD} \geq 2.7V$
Current Output I/O for 1 pin	$I_{IO(PIN)}$	-	-	20	mA	
Current Output I/O total ³	$I_{IO(SUM)}$	-	-	100	mA	
Analog Supply Voltage	V_{DDA}	1.62	-	3.6	V	
ADC Voltage Range	V_{ADC}	V_{SSA} ⁴	-	V_{DDA}	V	
Total Current through MCU	I_{MCU}	-	-	150	mA	

¹ Some pins on the chip are tolerant of incoming 5V, but it is strongly recommended to familiarise with the original datasheet of the chip before using this option.

² This specifications applies to most I/O pins and shows absolute values, but more data with different conditions is available in the datasheet of the chip.

³ This current consumption must be correctly distributed over all I/Os and control pins.

⁴ Analog Ground

Power Switches

In order to realize full control over the vehicle's functions, a system of high-side drivers was implemented. Nodes v1 used a high-side driver, which was not reliable enough for active use, so the new versions used the best component in terms of its characteristics at the time of writing, the VNQ9080AJ[11] (*IC1* and *IC2*), which proved to be quite reliable in many use scenarios. But starting from version 2.2.5 it is recommended to use VNQ9025AJ[16], which is almost exactly the same component as before, but has improved characteristics in terms of current and resistance. One such component is capable of providing 4 outputs, so to provide 8 outputs, two such components are used simultaneously, which are controlled using signals from the Power Node chip, which is mentioned in the previous Section Power Node. In general, this is the most important functionality that Power Node can provide, because the functionality and stability of the key parts of the car depend on it, so special attention should be paid when soldering, testing and using this functionality. As shown in Figure 15, resistors are connected to some of the driver pins. Their values and presence were chosen in strict accordance with the component datasheet and provide some protection against unexpectedly increased currents.

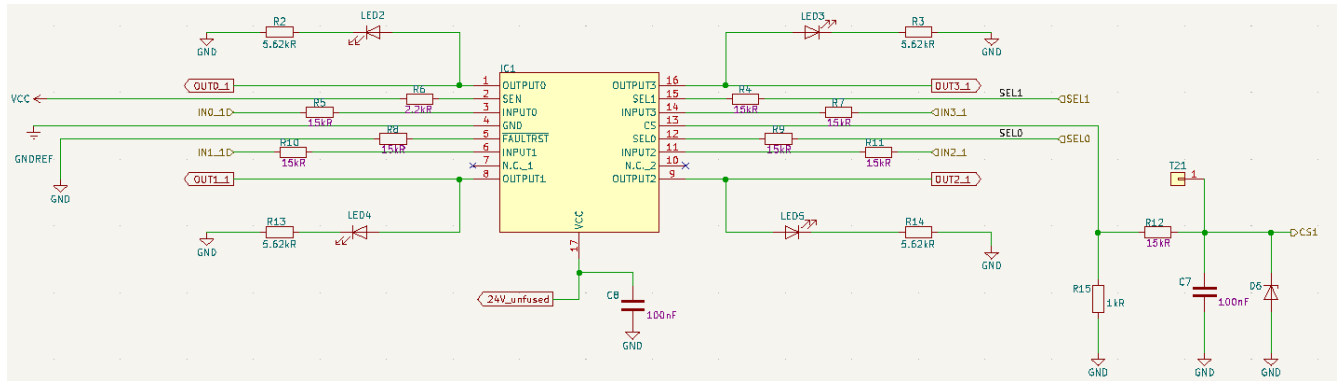


Figure 15: High-Side Driver Schematic

Table 2: High-Side Driver Pin Functions

Name	Function
V_{cc}	Power connection; The driver itself is connected to 24V, which immediately comes to the board, bypassing the fuse, because the high-side driver itself represents the functionality of the fuse. The fact that the datasheet of this component allows it to be used at higher voltages, as well as its parallel connection on the board, does not give permission to supply higher voltages. The maximum permissible voltage supplied to the board is mentioned in Table 7.
OUT_{0-3}	Power output
GND	Ground connection; The ground of this component is isolated in some sense from the rest using diode <i>D1</i> and resistor <i>R30</i> , as shown in Figure 16.
IN_{0-3}	Voltage controlled input pin; controls output switch state.
CS	Analog current sense output pin; it delivers a current proportional to the selected load current.
SEn	Enables the CS diagnostic pin.
$SEL_{0,1}$	Addresses the CS multiplexer.
$FaultRST$	Unlatches the output in case of fault - if kept low, sets the output to auto-restart mode.

¹ Some pins on the chip are tolerant of incoming 5V, but it is strongly recommended to familiarise with the original datasheet of the chip before using this option.

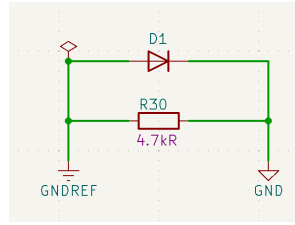


Figure 16: Ground Connection for the High-Side Driver

Each High-Side Driver output has its own LED that shows the power status. And each LED has its own color, so that in the field it is easier to navigate and understand what is responsible for what. More information about this can be found in Section LED Colour Description.

This High-Side Driver has a CS pin that sends a current signal proportional to the current at the output of a particular pin, which was selected as a result of interaction with the $SEL0$ and $SEL1$ pins (see Table 3 for the truth table).

Table 3: High-Side Driver Truth Table¹

Mode	IN_x	OUT_x	SEL_0	SEL_1	CS^3	Conditions
Normal ²	L	L	L	L	$I_{SENSE} = I_{OUT_0}/K$	Nominal load connected; $T_j < 150^\circ C$
			H	L	$I_{SENSE} = I_{OUT_1}/K$	
	H	H	L	H	$I_{SENSE} = I_{OUT_2}/K$	
			H	H	$I_{SENSE} = I_{OUT_3}/K$	
Overload	L	L	X	X	$V_{SENSE} = V_{SENSEH}$	Overload or short to GND causing: $T_j > T_{TSD}$ or $\Delta T_j > \Delta T_{j_SD}$
	H	H				
Undervoltage	X	L	X	X	Hi-Z	$V_{CC} < V_{USD}$ (falling)
OFF-state diagnostics	L	H	X	X	$V_{SENSE} = V_{SENSEH}$	Short to V_{cc} or Open-load
Negative output voltage	L	$< 0V$	X	X	Hi-Z	Inductive loads turn-off

¹ This table is a shortened version of the component truth table. FaultRST = L and SEN = H for all cases presented in this table, as dictated by the design decisions.

² The IN_x and OUT_x values are not related to the SEL_x and CS values.

³ The value of K is mentioned in the Table 5

By measuring the current I_{SENSE} on the CS pin, the current on each output pin I_{OUT} can be obtained, where the current value I_{SENSE} is K (from Table 5) times less than the current on the output pin I_{OUT} . That is, to calculate the output current on the selected pin, the Formula 4 can be used.

$$I_{SENSE} = I_{OUT_{0-3}}/K \quad (4)$$

But the chip that should read these changes can only measure the voltage V_{SENSE} , and not the current I_{SENSE} , Formula 5 shows the necessary and approximate calculation of Vsense using the product of the resistance R_{SENSE} ($R15$ or $R29$) and the current I_{SENSE} .

$$V_{SENSE} = R_{SENSE} * I_{SENSE} \quad (5)$$

But it is strongly recommended to keep in mind the maximum possible voltage on the CS pin V_{SENSEH} under Fault Condition (approximately 7.5V from Table 5), since the microchip itself wasn't designed for high values. So

it was decided in version 2.2.4 to add a zener diodes *D5* and *D6* to avoid overvoltages above 3.3V.

To measure the voltage, the V_{SENSE} chip uses a built-in 12-bit ADC, where each value from zero voltage to 3.3V is divided into 4096 points (from zero to 4095). That is, the calculation of the ADC value is performed according to Formula 6.

$$ADC_{value} = (V_{SENSE}/3.3) * 4095 \quad (6)$$

By combining Formulas 4, 5, 6, the current output I_{OUT} on the pin can be calculated based on the measured ADC value by Formula 7.

$$I_{OUT_{0-3}} = ADC_{value} * K * 3.3 / (R_{SENSE} * 4095) \quad (7)$$

When connecting any equipment to the outputs, it is worth remembering the total maximum permissible current limit I_{LIMH} supplied to the output pins from Table 5. If this value is exceeded, the component will go into protection and turn off as if it had overheated or reached the power limit. So it is necessary to correctly calculate the needs and balance the connection without overloading one component.

In case of replacing the High-Side Driver, the values of K and the maximum permissible current limit I_{LIMH} also change as a result, therefore, for a more accurate and safe calculation of the current consumed on the line, it is important to change the resistance of R_{SENSE} according to Formula 8 so that the measurements for the chip continue to be in the range from zero to 3.3V, which means for the current from zero to the maximum permissible value.

$$R_{SENSE} = 3.3 * K / I_{LIMH} \quad (8)$$

After this change, it is also necessary to update the code variables, which are the values of K and R_{SENSE} , so that the formula continues to calculate the current consumed on the line correctly with the new component.

But theoretically it is even possible not to change the code, but just change the resistance of R_{SENSE} , since in fact, when the component is changed, the ratio between K and Rsense changes, which affects the calculation of the ADC value, which means that if this ratio is kept constant with the new High-Side Driver can continue to use the same variables in the code as before. By combining this understanding of the constancy of the ratio and Formula 8, Formula 9 can be obtained, where it possible to calculate the appropriate resistance value for this specific case.

$$R_{SENSE_{new}} = R_{SENSE_{old}} * K_{new} / K_{old} = 3.3 * K_{new} / I_{LIMH_{old}} \quad (9)$$

But it is important to understand that when increasing the resistance, the V_{SENSE} voltage at maximum current also increases, which means that in the best case, the current values will be lost when calculating above the maximum value that was before, and in the worst case, this action can damage the chip due to increased voltage. And vice versa, when decreasing the resistance, the measurement accuracy decreases significantly, so the action of replacing the component and resistance without changing the variables in the code is not recommended and can be used only in the most extreme cases.

Table 4: Selecting and Calculating a Suitable R_{SENSE} Value

R_{SENSE}^1	Conditions
$R_{SENSE} = 3.3 * 5050 / 8 \approx 2083, 1\Omega^2$	Placement of component VNQ9025AJ[16] or replacement of component VNQ9080AJ[11] with this one with changing the necessary variables in the code.
$R_{SENSE} = 3.3 * 1500 / 4.2 \approx 1178.6\Omega^2$	Placement of component VNQ9080AJ[16] or replacement of component VNQ9025AJ[11] with this one with changing the necessary variables in the code.
$R_{SENSE} = 3.3 * 5050 / 4.2 \approx 3967.9\Omega^3$	Replacement of component VNQ9080AJ[11] with VNQ9025AJ[16] without changing the variables of previous component in the code.
$R_{SENSE} = 3.3 * 1500 / 8 \approx 618.8\Omega^3$	Replacement of component VNQ9025AJ[11] with VNQ9080AJ[16] without changing the variables of previous component in the code.

¹ Suitable values for variables are taken from Table 5

² According to the Formula 8

³ According to the Formula 9

All specifications are for $T_j = 25^\circ C$ unless otherwise noted.

If there is $>$ or $<$ symbol before the number, it means that for the case of $V_{cc} = 24V$ the value could be bigger or smaller respectively.

Table 5: High-Side Driver Characteristics

Parameter		Symbol	Min.	Typ.	Max.	Unit	Conditions
Supply Voltage		V_{cc}	4	24 ¹	28	V	
Undervoltage shutdown		V_{USD}	-	2.1	2.7	V	
On-state Resistance	[11]	R_{ON}	-	> 86	-	m Ω	$I_{OUT} = 0.7A$; $V_{cc} = 13V$
	[16]		-	> 25	-		$I_{OUT} = 2.1A$; $V_{cc} = 13V$
Current Limitation ²	[11]	I_{LIMH}	-	< 4.2	-	A	$V_{cc} = 22V$
	[16]		-	< 8	-		
Current Supply		I_S	-	> 5.5	7.5	mA	$V_{cc} = 13V$; $V_{SEn} = V_{FR} =$ $= V_{SEL} = 0V$; $V_{IN} = 5V$; $I_{OUT} = 0A$
Voltage Input Low ³		V_{INL}	-	-	0.9	V	
Voltage Input High ³		V_{INH}	2.1	-	-	V	
CS Voltage Output in Fault Condition		V_{SENSEH}	5	-	7.5	V	$13V < V_{cc} < 18V$
Ratio K (I_{OUT}/I_{SENSE})	[11]	K	-5%	1500	5%		$I_{OUT} = 2.1A$; $V_{SENSE} = 3.5V$; $V_{SEn} = 5V$
	[16]		-5%	5050	5%		$I_{OUT} = 6.3A$; $V_{SENSE} = 3.5V$; $V_{SEn} = 5V$
Shutdown temperature		T_{TSD}	150	175	210	$^\circ C$	
Dynamic temperature		ΔT_{j_SD}	-	< 55	-	$^\circ C$	$V_{cc} = 19V$

¹ This number is not dictated by the datasheet, but by the use of the component in Nodes itself. In the datasheet, a typical number is 13V.

² $I_{LIMH} = MAX(I_{OUT})$

³ This is also true for the SEL and SEn pins.

PWM

In order to control the power of connected equipment, such as fans or pumps, using a digital signal, the Nodes have a PWM (Pulse-width modulation) function implemented on two pins, which, as a result of changing the duty cycle, affects the operation of the equipment using just one connected signal. The larger the duty cycle, the greater the power consumption (for example, a fan rotates faster with a larger duty cycle). It is also important to comply with the permissible PWM signal frequencies for connected devices, which can be found in their datasheets.

Some devices require a PWM signal of greater amplitude, while the chip on the Power Node can only output 3.3V amplitude of the required signal. Therefore, starting with version 2.2.2, Level-Shifters are actively used as shown in Figure 18. The way the Level Shifter works in this setup can be confusing, but there is an explanation for its functionality. Due to the lack of ground in this circuit, the supplied low signal pulls the current from the high voltage to itself and as a result, almost zero is formed on the output signal, because the current chooses the path of least resistance to the ground, and since Level-Shifter consists of a MOSFET transistor TN2106[13] ($Q1$ and $Q2$), it does not pass through itself a voltage higher than 3.3V, since this voltage is connected to the gate. If the input signal is high, then the current of higher voltage does not go towards the transistor, because now the path is closed for it and as a result, a high signal of increased amplitude is formed at the output.

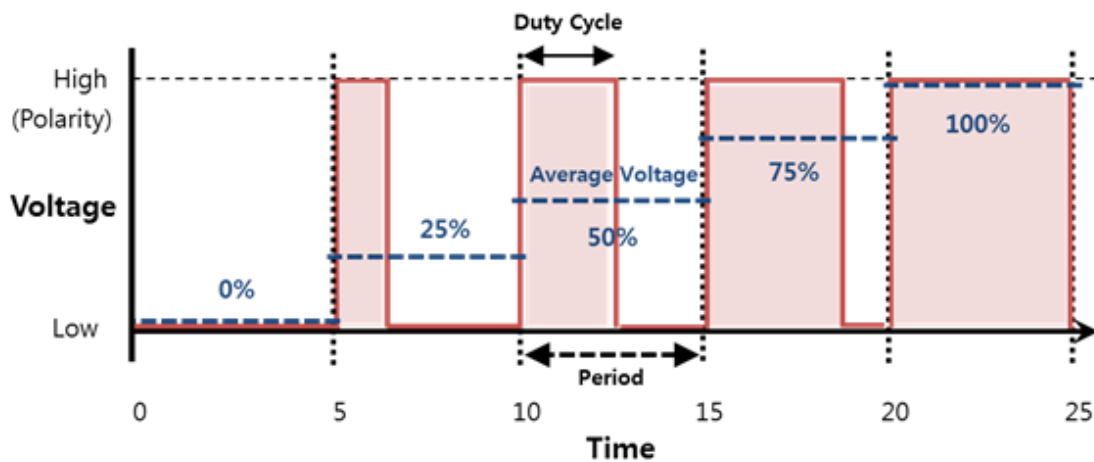


Figure 17: PWM Waveforms[12]

Starting with version 2.2.3, both pins are capable of providing an increased controlled amplitude, while 2.2.2 can only provide 24V on one PWM1. In version 2.2.3, it became possible to change the amplitude of the output signal as needed using a small switch and easily select one of three modes (3.3V, 5V and 24V), which provides greater versatility and ease of use and testing.

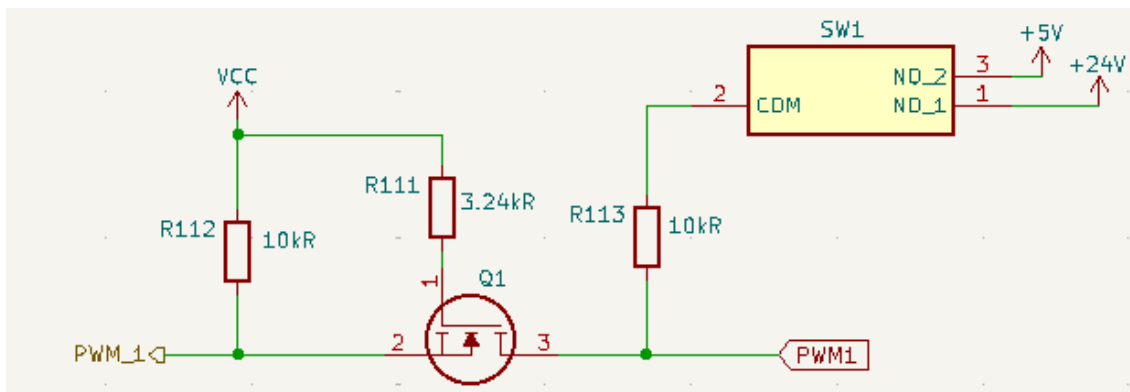


Figure 18: PWM Level-Shifter

Analog Node

Another equally important part of Nodes is the Analog Node. The Analog Node chip acts as a processor of filtered signals received from connected sensors, and then sends the received data further via CAN. The Analog Node power is supplied separately from the main 3.3V power supply to ensure maximum stability and safety of this part, which is critical for the analog signal processing. The Analog Node power is controlled using the Power Node, as described in Section Controllable Power for Analog Node. In version 2.1, the controlled power is supplied from the switch, and in version 2.2, it is supplied from the voltage regulator.

Both versions have 16 ADC inputs with different amplitudes (3.3V; 5V and 24V) that can handle incoming analog signals (more information about this in Section Filters), as well as three I2C inputs that can accept digital signals from connected sensors, which increases the versatility of the board and allows for a wider range of use cases.

There are also two controllable LEDs (*LED10* and *LED11*) that can be used to monitor the operation of the chip and certain functions. More information about the LEDs can be found in Section LED Colour Description.

This section contains information that differs from the information described earlier, so first of all it is worth looking at Section Power Node, which also describes the basic operating principles of the STMicroelectronics chip and the nuances associated with it, such as JTAG, BOOT pin and others.

Analog Node v2.1

The Analog Node version 2.1 uses the STM32H7B0RBT6[14] (*U2*) chip, which has two CANs, but only one of them was actively used. Unfortunately, the chip did not show its best side in practical use, namely insufficient reliability and price.

This version makes active use of the external ASE-25.000MHZ-LC-T[15] (*Y2*) oscillator, which provides a stable and recommended frequency of 25 MHz according to the Table 6.

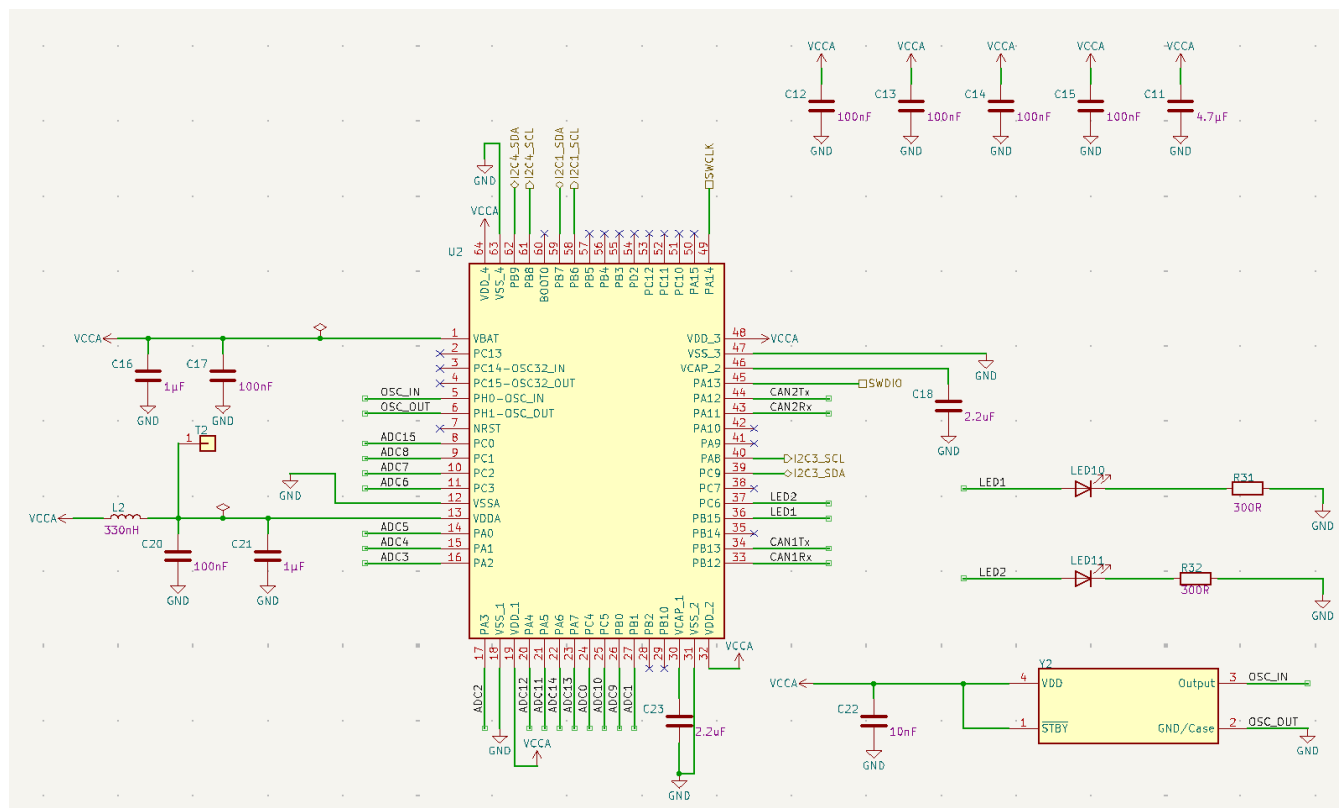


Figure 19: Analog Node schematic v2.1

Table 6: Analog Node v2.1 Chip[14] Characteristics

Parameter	Symbol	Min.	Typ.	Max.	Unit	Conditions
Supply Voltage	V_{DD}	1.62	3.3	3.6	V	
External Oscillator Frequency	f_{OSC_IN}	4	25	50	MHz	
Current Consumption	$I_{c(MCU)}$	?	?	?	mA	
Voltage Input Low I/O	V_{IL}	-	-	$0.3 * V_{DD}$	V	$1.62V < V_{DD} < 3.6V$
Voltage Input High I/O ¹	V_{IH}	$0.7 * V_{DD}$	-	-	V	
Voltage Output Low I/O ²	V_{OL}	-	-	1.3	V	$ I_{IO} = 20mA$
Voltage Output High I/O ²	V_{OH}	$V_{DD} - 1.3$	-	-	V	$V_{DD} \geq 2.7V$
Current Output I/O for 1 pin	$I_{IO(PIN)}$	-	-	20	mA	
Current Output I/O total ³	$I_{IO(SUM)}$	-	-	140	mA	
Analog Supply Voltage	V_{DDA}	1.62	-	3.6	V	
ADC Voltage Range	V_{ADC}	V_{SSA} ⁴	-	V_{DDA}	V	
Total Current through MCU	I_{MCU}	-	-	620	mA	

¹ Some pins on the chip are tolerant of incoming 5V, but it is strongly recommended to familiarise with the original datasheet of the chip before using this option.

² This specifications applies to most I/O pins and shows absolute values, but more data with different conditions is available in the datasheet of the chip.

³ This current consumption must be correctly distributed over all I/Os and control pins.

⁴ Analog Ground

Analog Node v2.2

In this version, Analog Node has acquired a new chip STM32G431RBT6[8] (*U11*) of the same group as the chip from Power Node, which required code adaptation, but allowed using a cheaper, more reliable and time-tested chip compared to what was before. Also, as a result of changing the chip, only one CAN line remained, which optimized and simplified the board in use. The oscillator is now exactly the same as on Power Node and produces 8 MHz, which is still sufficient. In general, this is the same chip as for Power Node, but the selected chip has a larger number of pins and more possible connections. More information about the chip can be found in Section Power Node.

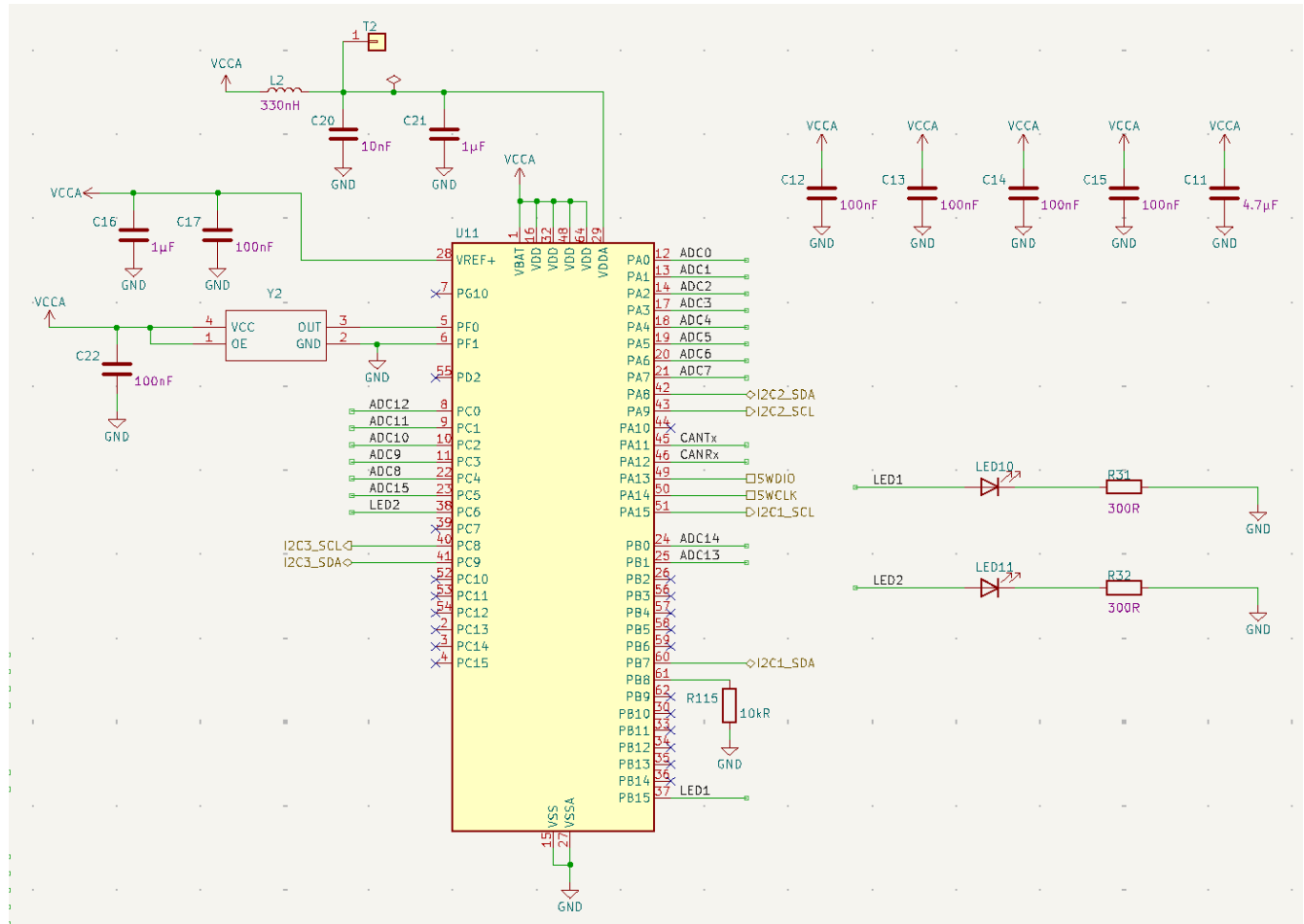


Figure 20: Analog Node schematic v2.2

Filters

Filters are a very important part of Analog Node, because vital signals from sensors pass through them to have more stable and readable data. The filtering system consists of several important elements, namely an active low-pass filter, consisting of a simple and understandable op-amp LM358[17], which is intelligently combined with a passive low-pass filter, consisting of RLC components, which ensures maximum versatility and quality of signal filtering, and a voltage divider, which in combination with a capacitor also forms a low-pass filter. This system both filters the signal actively and passively, and also reduces the signal amplitude to readable for the chip 3.3V, and it is absolutely unique in its simplicity and ease of use, as well as the quality provided, so as a result of which, starting from the Nodes old version v1, the design remains unchanged to this day.

In total, there are 16 possible connections, but two of them are allocated for a signal with an amplitude of 3.3V (Figure 21), seven filters for 5V (Figure 22), three filters for a signal with an amplitude of 24V (Figure 23) and the remaining 4 are allocated for the future for the idea of conveniently connecting additional filters, which is still at the concept stage, so it can be removed in the future.

If for some reason the sensors produce a non-standard signal amplitude, which was not initially included in the Nodes, then the resistance on the voltage divider needs to be recalculated with Formula 10, where R_L is resistor connected to the ground (for example $R99$ from the Figure 22) and R_H connected to the filter (for example $R96$ from the Figure 22).

$$V_{OUT} = V_{IN} * R_L / (R_L + R_H) \quad (10)$$

Formula 10 could be simplified into Formula , because $R_L = 3.24k\Omega$ and $V_{OUT} = 3.3V$ in most cases.

$$R_H = 3.24 * 10^3 * (V_{IN} / 3.3 - 1) \quad (11)$$

The supplied voltage should not be higher than the powered voltage, namely 24V, to avoid breakdowns and problems with op-amp.

Each output has a testpoint (for example $T20$) that can be used for testing, but more information about this can be found in Section Testpoints Description.

If the supplied signal amplitude is greater than 9V, then it is necessary to use a 1nF capacitance instead of 10nF capacitors like on the Figure 23.

Also, if an NTC sensor is connected, it is important to short the inductor (for example $L25$ from the Figure 22) and the resistor (for example $R95$ from the Figure 22) on its path, and solder a 680 ohm resistor (for example $R98$ from the Figure 22) into the DNA position, but this action requires subsequent tests after that.

When a 5000 step resolution is wanted and lower offset voltage, then it is recommended to use ADA4522-2 or more expensive solutions of the same footprint SOIC-8 instead of LM358[17].

More theoretical information about the operation of filters can be found in the excellent Scott's thesis[18]

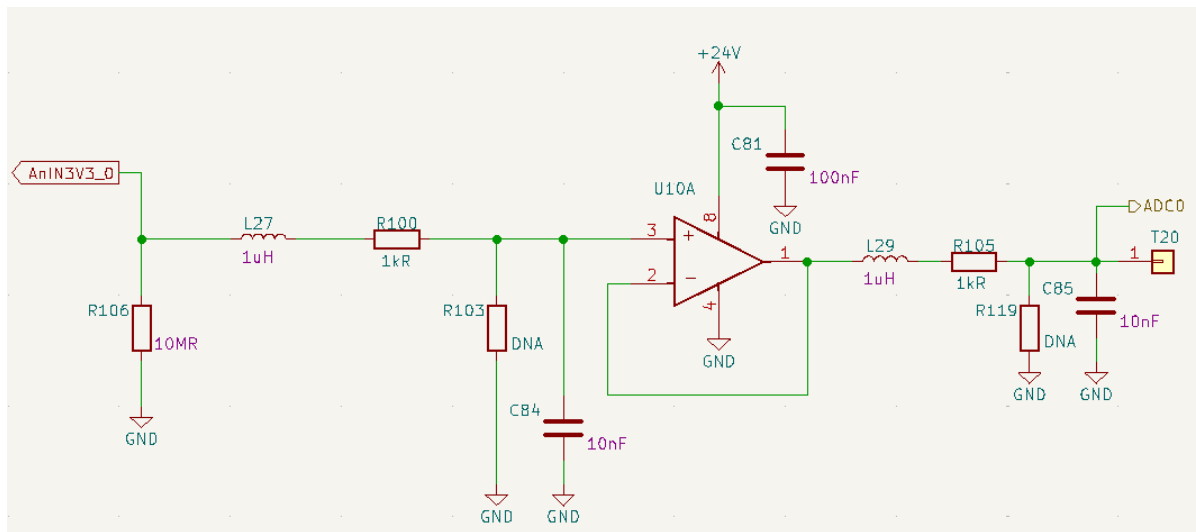


Figure 21: Filtering System for 3.3V amplitude signal

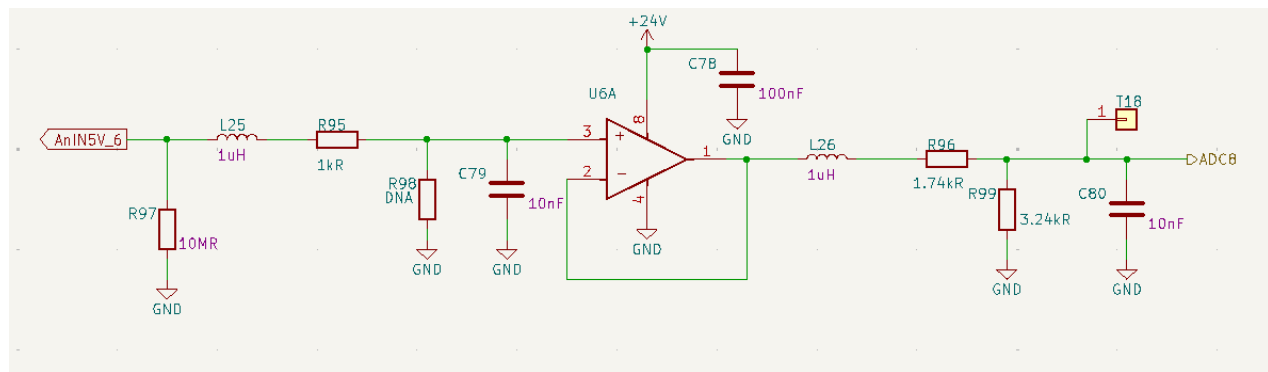


Figure 22: Filtering System for 5V amplitude signal

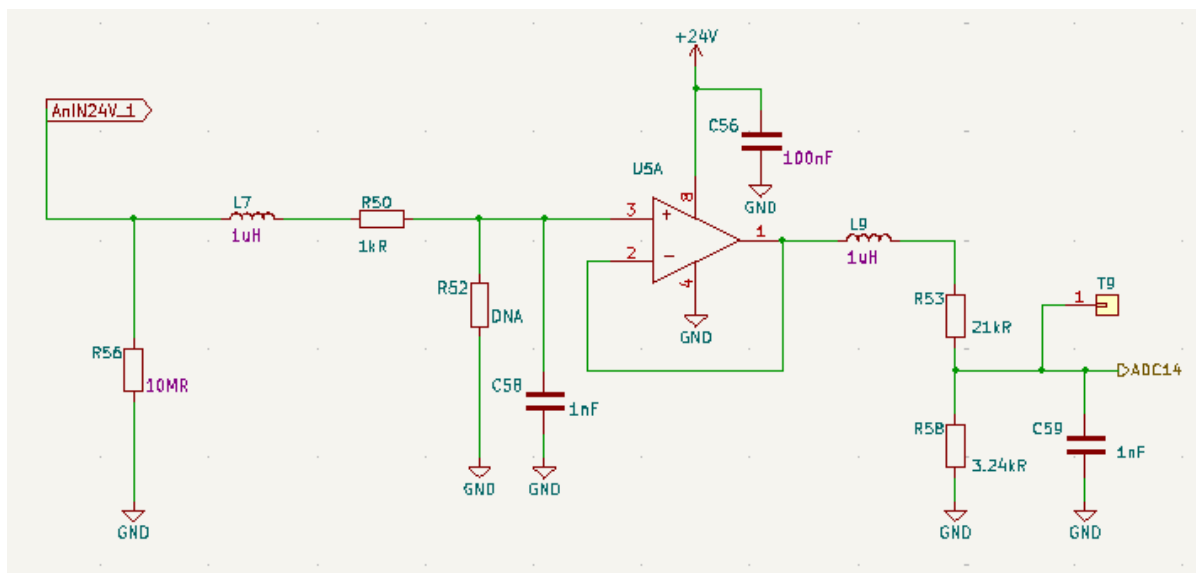


Figure 23: Filtering System for 24V amplitude signal

CAN

In order for the communication between Nodes and ECU to be as stable as possible, CAN is used, which, thanks to two connection lines (High and Low), is able to work in a variety of conditions, which is important during active operation of the car. This protocol is quite reliable, simple thanks to the priority system and protected from external interference. In version 2.1, Nodes had two different CANs, but in the new versions 2.2 there is only one left, since the Analog Node chip was replaced. The chips themselves that transmit and receive data are not able to directly send data to the main CAN line, so auxiliary components TCAN1462VDRQ1[19] (*IC3* and *IC4*) are used that were specially designed for this. These components are powered by 5V, so if ST-LINK is connected, but not the main 24V power supply, CAN will not work due to insufficient voltage.

In order for data to be exchanged on the main line, a terminating resistance is needed between High and Low. There must be two terminators at different ends of the main data transmission line and one of them is located on Nodes. The resistance value should be about 120 Ohms on one terminator. In new versions, a switch *SW3* was introduced that allows to connect or disconnect the terminator without soldering, which saves time. In general, just two 60 Ohm resistors connected in series are enough, as it was before, but in new versions, together with the switch, there are now 4 resistors (*R46*, *R37*, *R47* and *R109*) instead of two. This is necessary in case of switching off the terminator, data transmission will not be affected, so it simply bypasses these resistors along the least resistance path.

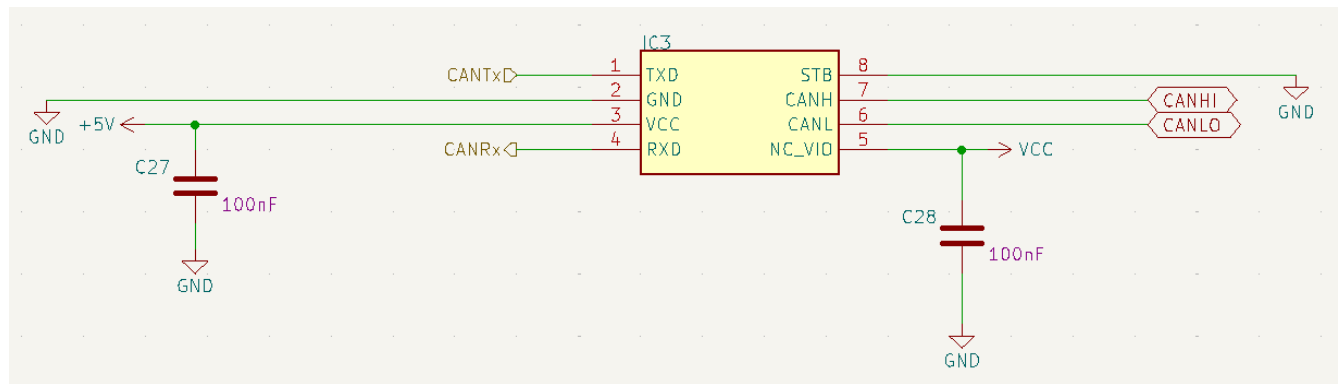


Figure 24: CAN schematic for Power Node

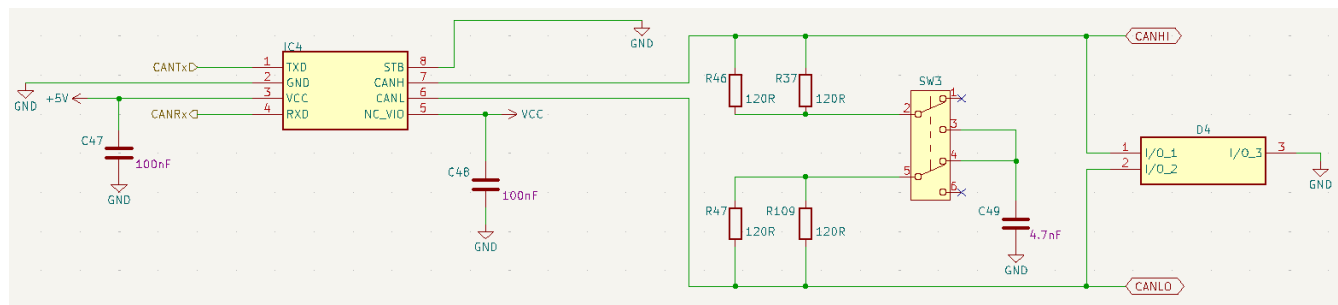


Figure 25: CAN schematic for Analog Node

Electrical Specifications

At the time of writing, this section contains mostly theoretical data that was obtained based on the documentation of the components used on the board.

All specifications are in $10^{\circ}C \leq T_j \leq 50^{\circ}C$ unless otherwise noted.

Table 7: Electrical Characteristics

Parameter		Symbol	Min.	Typ.	Max.	Unit	Conditions
Forward Voltage $D2^1$	v2.1	V_f	-	500 ^[2]	-	mV	
	v2.2		-	370 ^[3]	-		
Voltage Input ²		V_{in}	$22.8+V_f$	$24.0+V_f$	$25.2+V_f$	V	
Current Consumption	v2.1	I_c	?	?	?	mA	
	v2.2		?	?	?		
Current Output ³	v2.1	I_o ⁴	-	$350-I_c$	-	mA	$V_o = 3.3V^{[4]}$
			-	$350-I_c$	-		$V_o = 5V^{[4]}$
	v2.2		-	$300-I_c$	-		$V_o = 3.3V^{[5]}$
			-	$1000-I_{o_{3v3}} - I_c$ ⁵	-		$V_o = 5V^{[6]}$

¹ V_f is the standard forward voltage of the diode $D2$ connected in series. So if just 24V are connected to the Nodes it means that the board itself after diode ($D2$) will receive $24 - V_f$, which is not critical in theory, but still should be taken into account.

² Based on the datasheet of Zener Diode ($D3$)^[1].

³ The values in this section of the table represent the theoretical maximum current output that a particular output voltage pin can provide on different versions. The values are based on voltage regulators' datasheets, so if your use case requires higher current, it is highly recommended to familiarise with the limitations and assumptions of the voltage regulator on a particular line.

⁴ In order to at least roughly understand the available current on the line, due to the current consumption of the board itself, it is necessary to calculate available current for the equipment by using presented equation. This calculation does not provide the most accurate value of the output current, since the I_c is the sum of all consumed currents on the board itself (use the maximum value for the calculation from the table for each version or measure by yourself current consumption of the board before an active usage); accordingly, the calculation takes place with a small margin for the board itself.

⁵ To perform this calculation, It is needed to know the output current $I_{o_{3v3}}$ on the line 3.3V, since the 5 volt and 3.3 volt voltage regulators are connected in series in version 2.2, which may affect the current output value, so this fact must be taken into account when connecting additional equipment.

Absolute Maximum Ratings

This section could be updated later, when the team will have more spare boards to test.

Table 8: Absolute Maximum Ratings

Parameter	Symbol	Rating		Unit
		v2.1	v2.2	
Maximum allowed current consumption	I_{max}	1 ¹		A
Maximum repetitive peak reverse voltage	V_{RRM}	25 ^[2]	30 ^[3]	V

¹ This value is dictated by the choice of the fuse. Since the board itself has a connection for the most popular type of fuses, namely blade fuses, the minimum value that is on the market is 1A, but it is strongly recommended to use 0.5A fuses if they will be available to buy, since in many current consumption tests Nodes did not show too high values.

Note: Stresses above those listed under Absolute Maximum Ratings can cause permanent damage to the device. This is a stress rating only. The functional operation of the device is not implied in any conditions above those indicated in the Electrical Specifications section.

Pin Functions

Table 9: Nodes Pin Functions

Name	Function
+3V3	Voltage outputs 3.3V obtained from voltage regulator. ¹
+5V	Voltage outputs 5V obtained from voltage regulator. ¹
+24V	Voltage outputs 24V after fuse and zener diode. ¹
PWR_{0-3}	Voltage outputs after an additionally connected filter. ²
+24V _{IN}	Power supply 24V for Nodes. ¹
GND	Ground connection.
AnIN3V3 ₀₋₁	Analog inputs for 3.3V signal amplitude. ²
AnIN5V ₀₋₆	Analog inputs for 5V signal amplitude. ²
AnIN24V ₀₋₂	Analog inputs for 24V signal amplitude. ²
AnIN ₀₋₃	Analog inputs for additionally connected filter. ²
PWM ₁₋₂	PWM outputs with different signal amplitudes. ³
OUT0 – 3 ₁₋₂	Controllable High-Side Driver outputs 24V. ⁴
I2C(1 – 3) _{SCL-SDA}	Input for digital signals of the I2C protocol. ⁵
CANHI and CANLO	Connection for CAN. In version 2.1 there are two CANs, but CAN2HI and CAN2LO are the main ones, like in version 2.2, since on this line there is a connection for both Power Node and Analog Node, while CAN1HI and CAN1LO have a connection only for Analog Node. ⁶

¹ More information in the Sections Electrical Specifications and Power Supply.

² More information in the Section Filters.

³ More information in the Section PWM.

⁴ More information in the Section Power Switches.

⁵ More information in the Section Analog Node.

⁶ More information in the Section CAN.

LED Colour Description

This section lists the LEDs on the board and their functions. This may not match the board in practice, as other LEDs may be soldered, so it is recommended to double-check which colors and LEDs are actually on the board before testing and usage.

Table 10: LED Description

Reference	Colour	Description
<i>LED1</i>	Yellow	Power Node Controlled LED ¹
<i>LED2</i>	Red	High-Side Driver's LED connected to OUT0.1 ²
<i>LED3</i>	White	High-Side Driver's LED connected to OUT3.1 ²
<i>LED4</i>	White	High-Side Driver's LED connected to OUT1.1 ²
<i>LED5</i>	Red	High-Side Driver's LED connected to OUT2.1 ²
<i>LED6</i>	Red	High-Side Driver's LED connected to OUT0.2 ²
<i>LED7</i>	White	High-Side Driver's LED connected to OUT3.2 ²
<i>LED8</i>	White	High-Side Driver's LED connected to OUT1.2 ²
<i>LED9</i>	Red	High-Side Driver's LED connected to OUT2.2 ²
<i>LED10</i>	Green	1st Analog Node Controlled LED ³
<i>LED11</i>	Orange	2nd Analog Node Controlled LED ³
<i>LED12</i>	Red	+24V ⁴
<i>LED13</i>	Yellow	+3.3V ⁴
<i>LED14</i>	Green	+5V ⁴
<i>LED15</i>	Blue	+3.3V for Analog Node ⁵

¹ More information in the Section Power Node.

² More information in the Section Power Switches.

³ More information in the Section Analog Node.

⁴ More information in the Section Power Supply.

⁵ More information in the Section Analog Node.

⁶ More information in the Section Controllable Power for Analog Node.

Testpoints Description

Testpoints allow to measure voltage in a variety of places and check the board's functionality. In most cases, to measure voltage on Nodes it is necessary to connect the negative probe to T6, which is the ground connection, and the positive probe to any other testpoint to measure the voltage at it. Table 11 lists all testpoints.

Table 11: Testpoints Description

Reference	Description
T1	The testpoint is used to measure voltage on the analog power supply pin <i>VDDA</i> of Power Node. ¹
T2	The testpoint is used to measure voltage on the analog power supply pin <i>VDDA</i> of Analog Node. ²
T3	The testpoint is used to measure voltage across the resistor <i>R33</i> together with the test point <i>T4</i> to find the current consumption of Nodes. ³
T4	The testpoint is used to measure 24V and also, together with <i>T3</i> , to measure current. ³
T5	The testpoint is used to measure the voltage on the 3.3V power supply line. ³
T6	Ground testpoint.
T7	The testpoint is used to measure the voltage on the 5V power supply line. ³
T8	The testpoint is used to measure the voltage on the 3.3V power supply line for Analog Node and also, together with <i>T23</i> , to measure current on this line. ⁴
T9	The testpoint is used to measure the filtered signal on the Analog Input for 24V #1. ⁵
T10	The testpoint is used to measure the filtered signal on the Analog Input for 24V #2. ⁵
T11	The testpoint is used to measure the filtered signal on the Analog Input for 24V #0. ⁵
T12	The testpoint is used to measure the filtered signal on the Analog Input for 5V #0. ⁵
T13	The testpoint is used to measure the filtered signal on the Analog Input for 5V #1. ⁵
T14	The testpoint is used to measure the filtered signal on the Analog Input for 5V #2. ⁵
T15	The testpoint is used to measure the filtered signal on the Analog Input for 5V #3. ⁵
T16	The testpoint is used to measure the filtered signal on the Analog Input for 5V #4. ⁵
T17	The testpoint is used to measure the filtered signal on the Analog Input for 5V #5. ⁵
T18	The testpoint is used to measure the filtered signal on the Analog Input for 5V #6. ⁵
T19	The testpoint is used to measure the filtered signal on the Analog Input for 3.3V #1. ⁵
T20	The testpoint is used to measure the filtered signal on the Analog Input for 3.3V #0. ⁵
T21	The testpoint is used to measure the voltage on the CS pin of high-side driver #1. ⁶
T22	The testpoint is used to measure the voltage on the CS pin of high-side driver #2. ⁶
T23	The testpoint is used to measure voltage across the resistor <i>R35</i> together with the test point <i>T8</i> to find the current consumption of the Analog Node. ⁴

¹ More information in the Section Power Node.

² More information in the Section Analog Node.

³ More information in the Section Power Supply.

⁴ More information in the Section Controllable Power for Analog Node.

⁵ More information in the Section Filters.

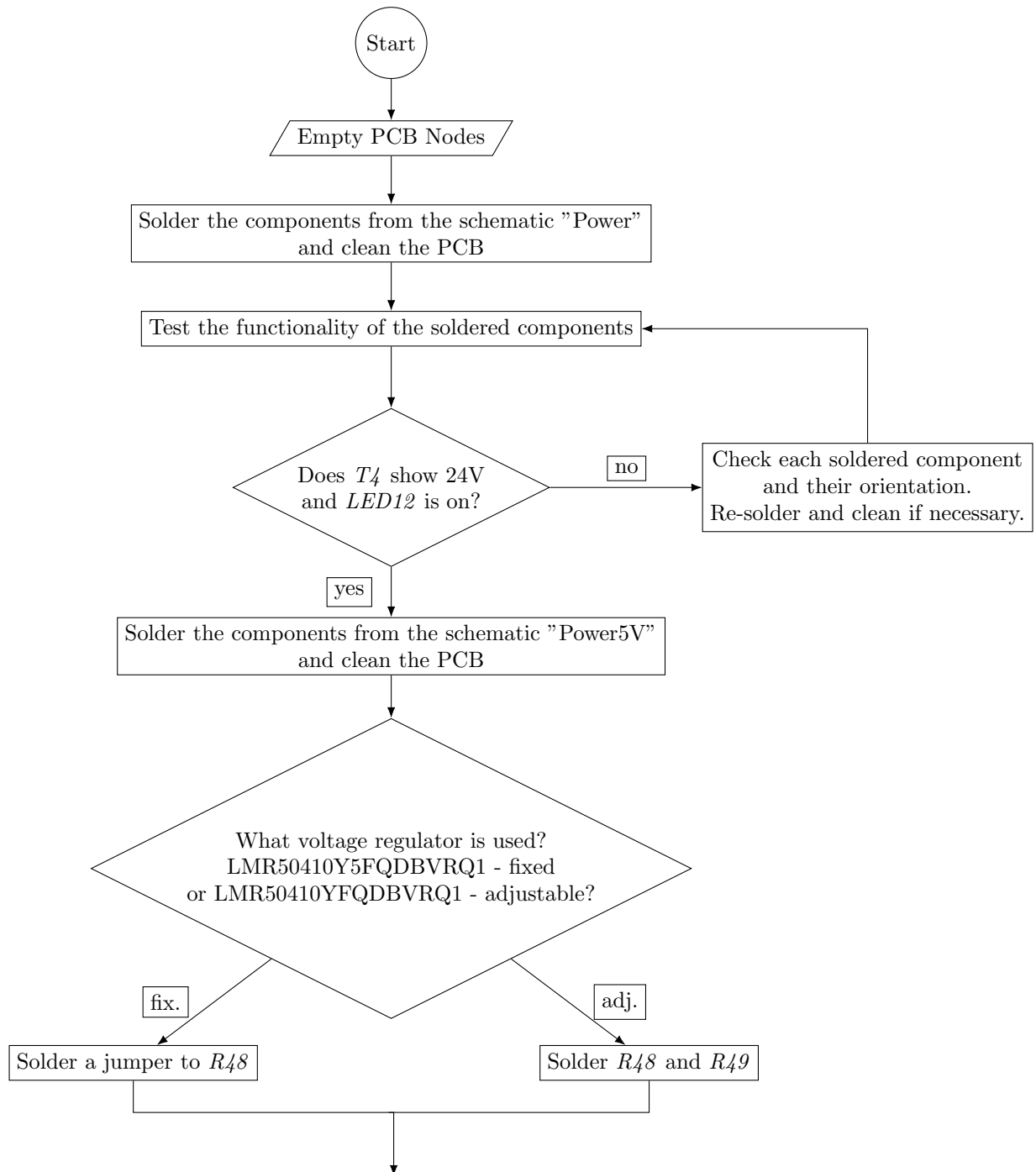
⁶ More information in the Section Power Switches.

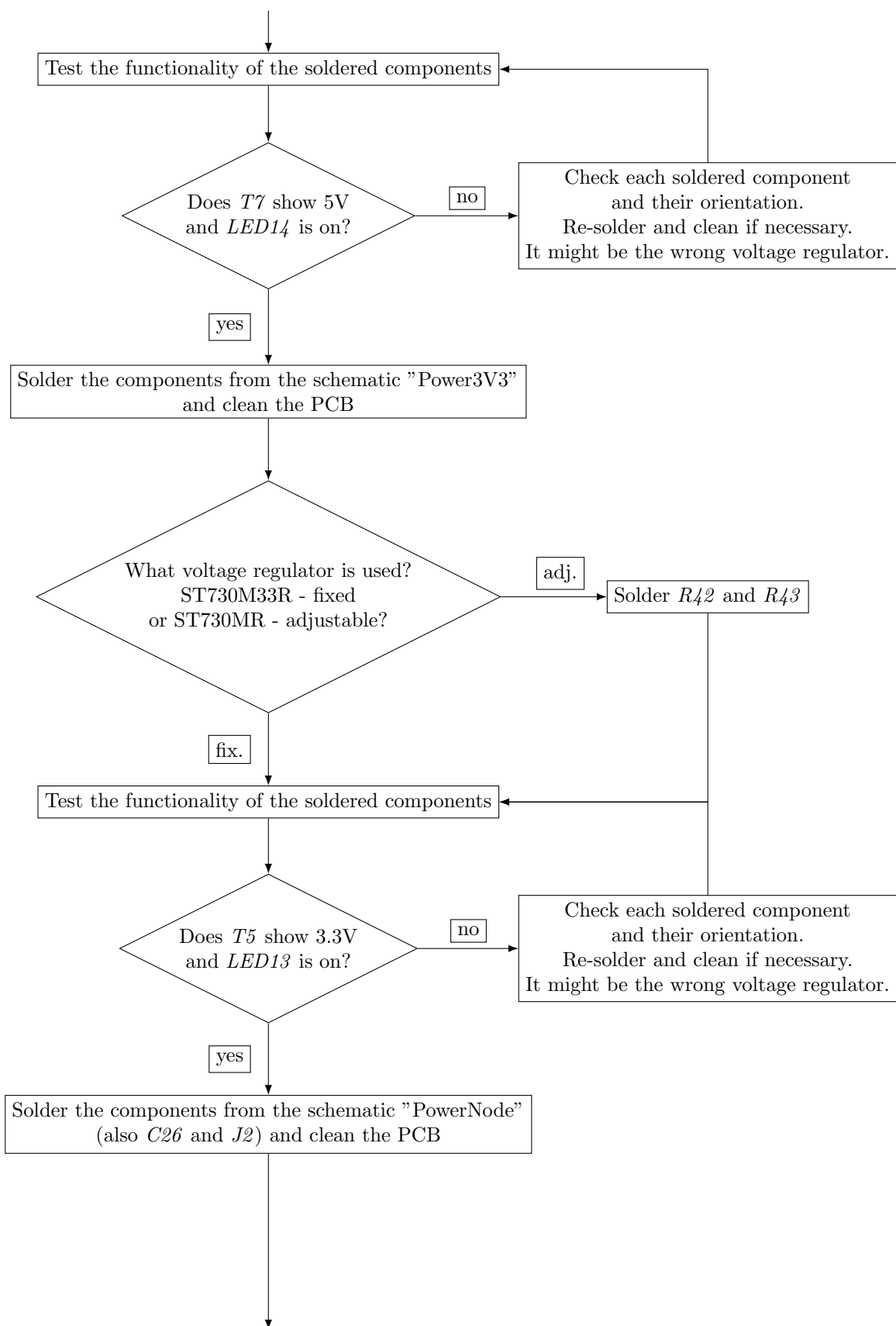
Usage

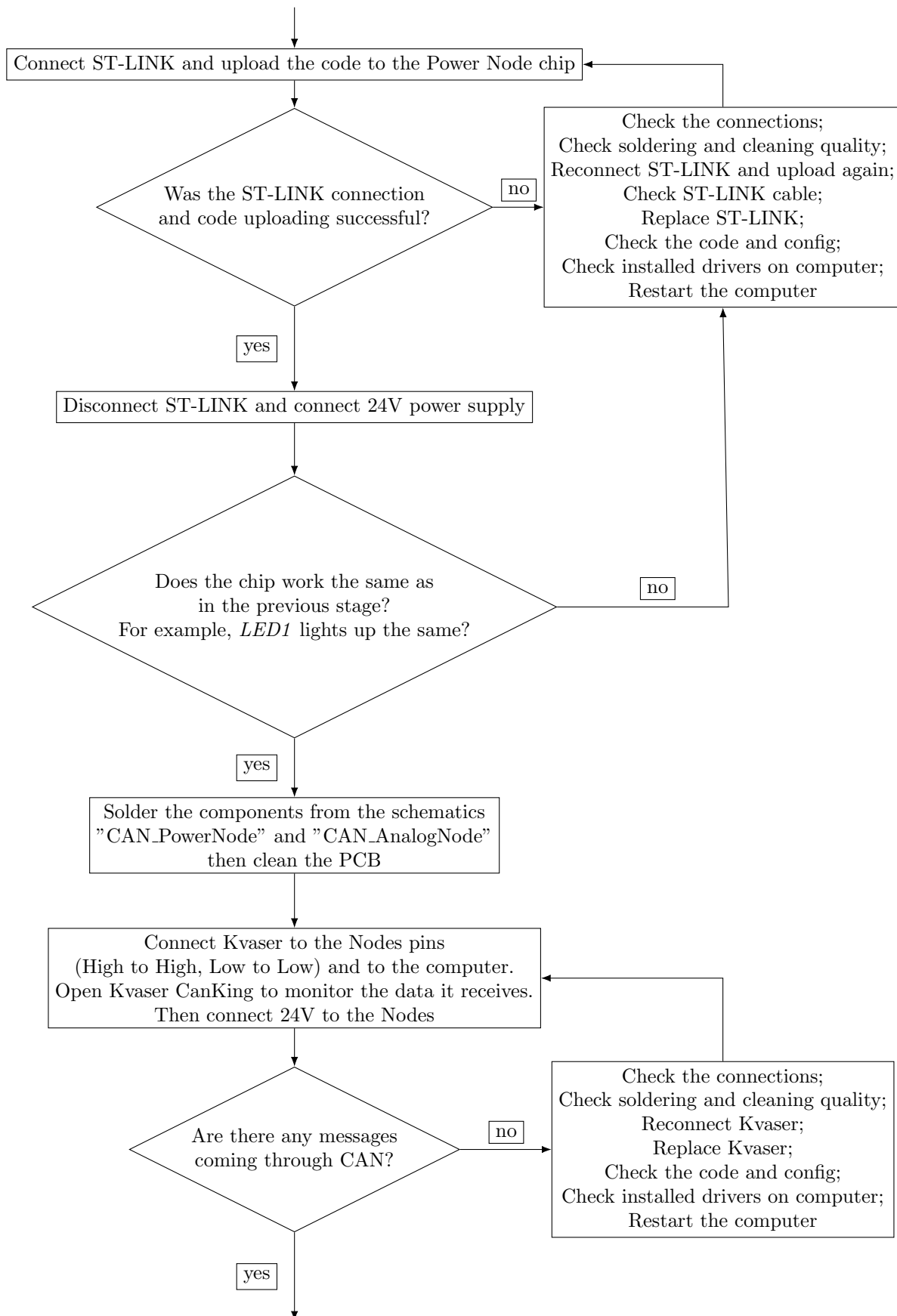
Explain here how to operate the system. Provide a schematic of the system’s operating logic.

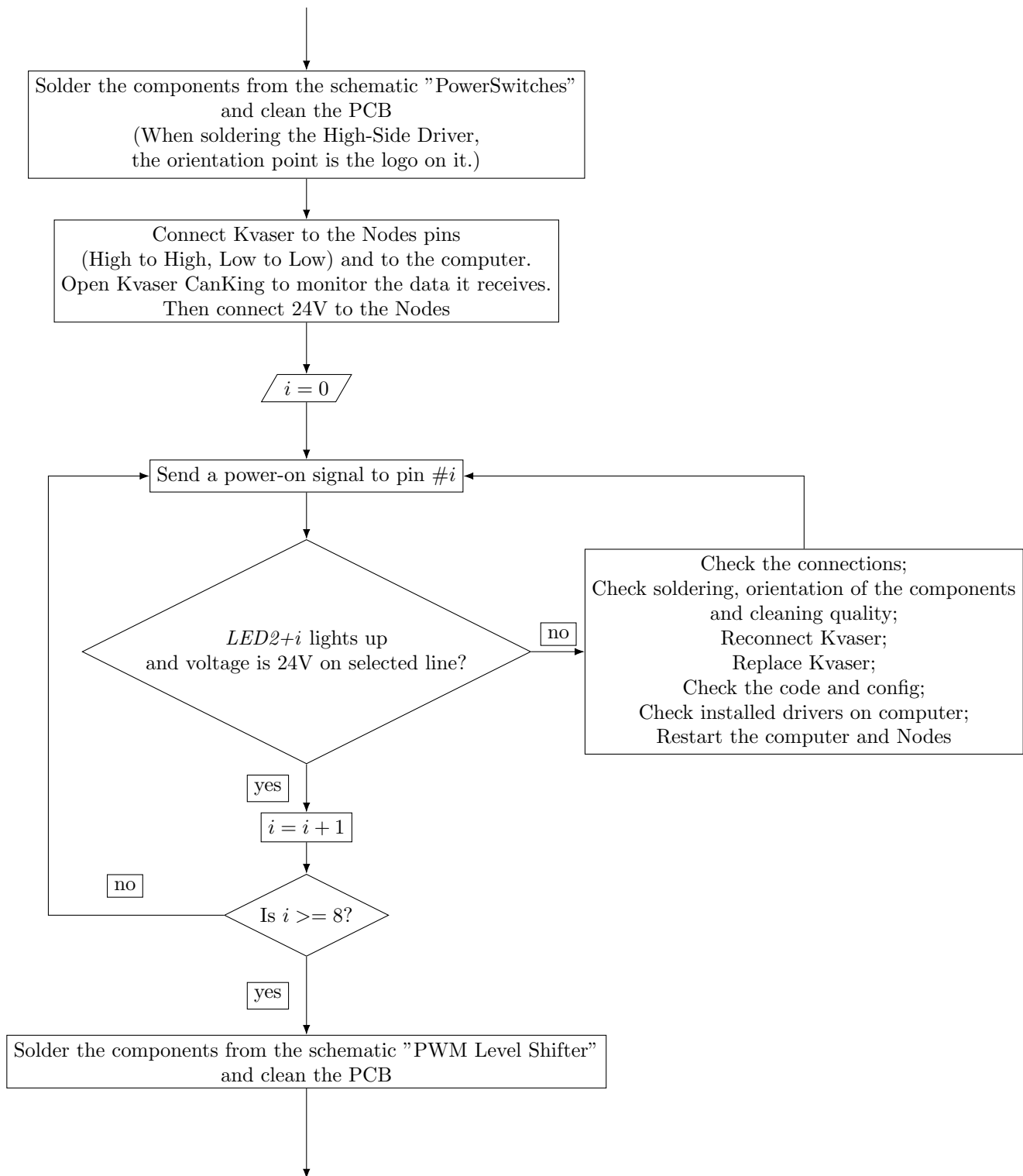
Assembly

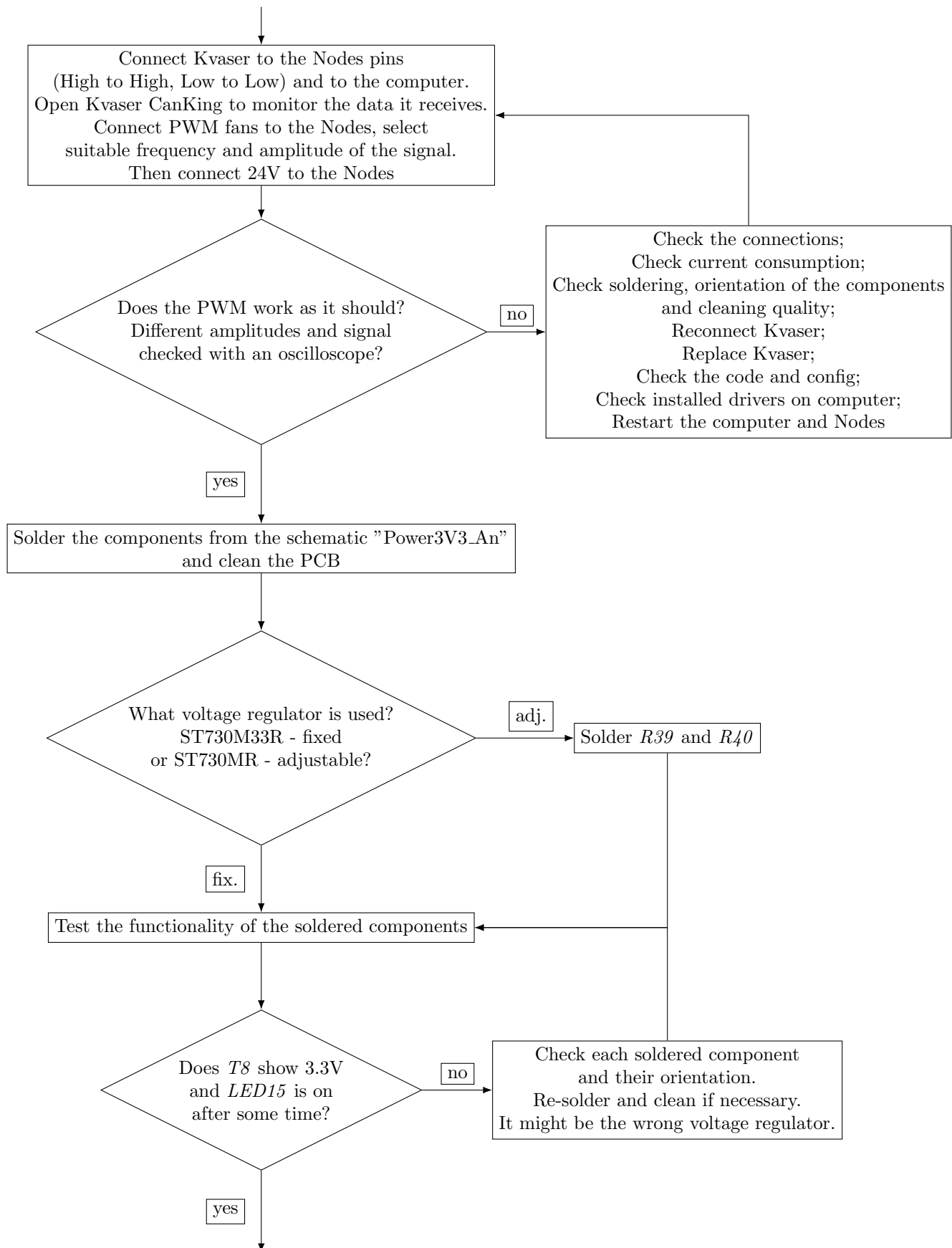
In order for Nodes to work as stably as possible after assembly, it is necessary to adhere to the following algorithm for installing components, where after each stage, the board must be thoroughly cleaned and tested. This instruction may differ slightly due to different components and boards, but in a general sense this algorithm will work if the exact meaning of some step is clear. Particular attention during testing should be paid to the board's power supply, because if any error is made at this stage, then the stable operation of the remaining components is not guaranteed. The instruction is suitable for version 2.2 Nodes and newer. Maintain the correct soldering iron temperature according to the component documentation.

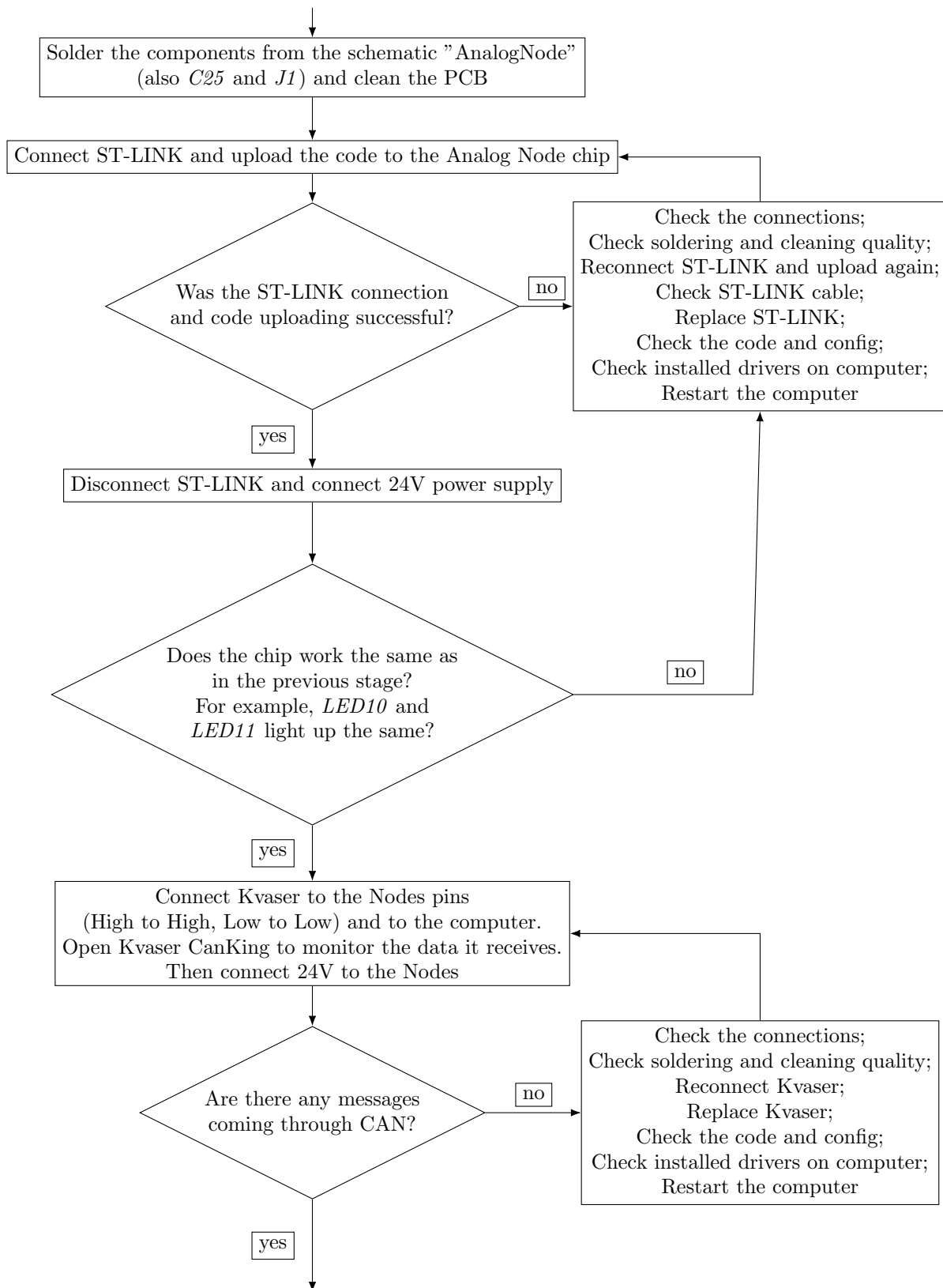


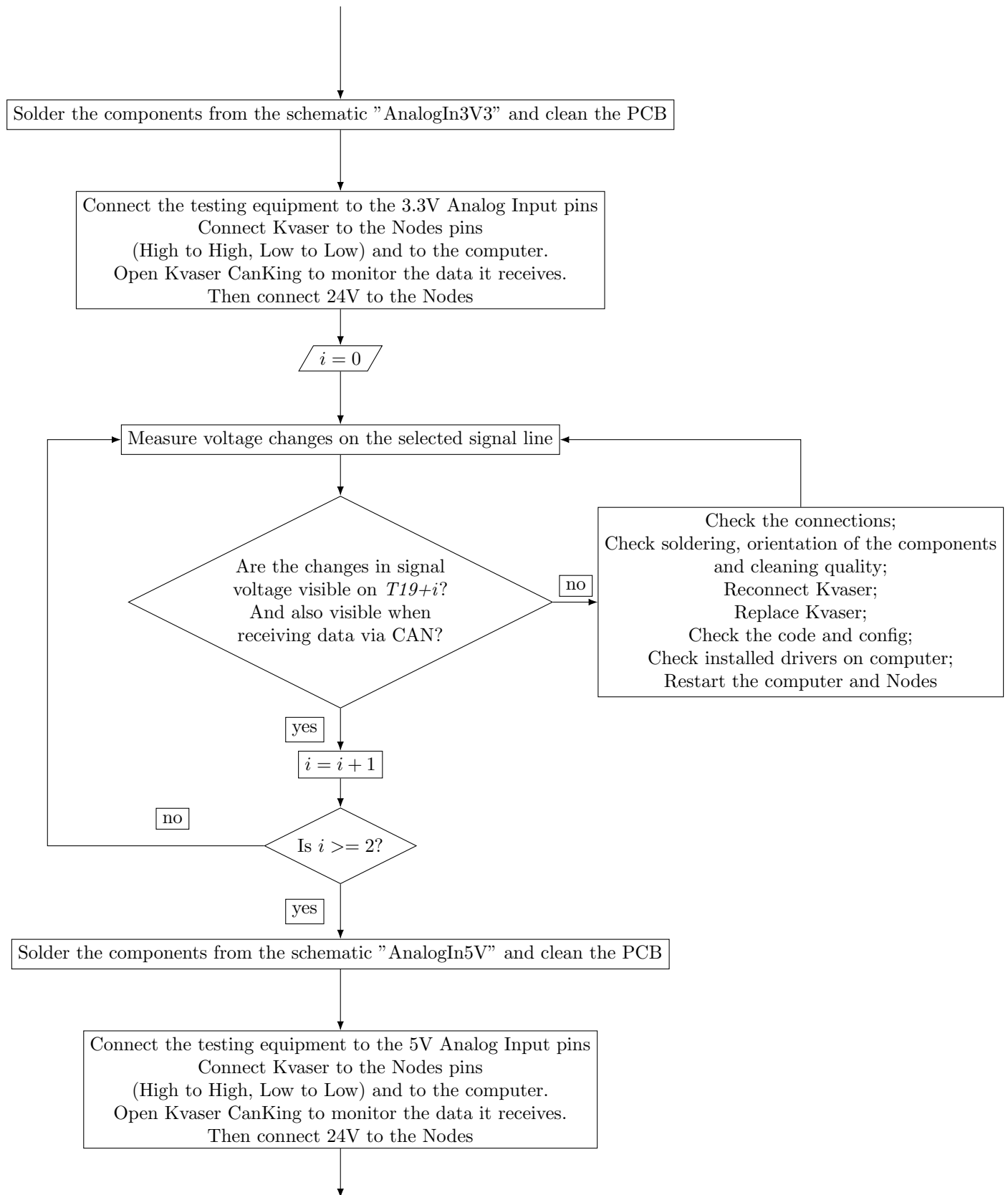


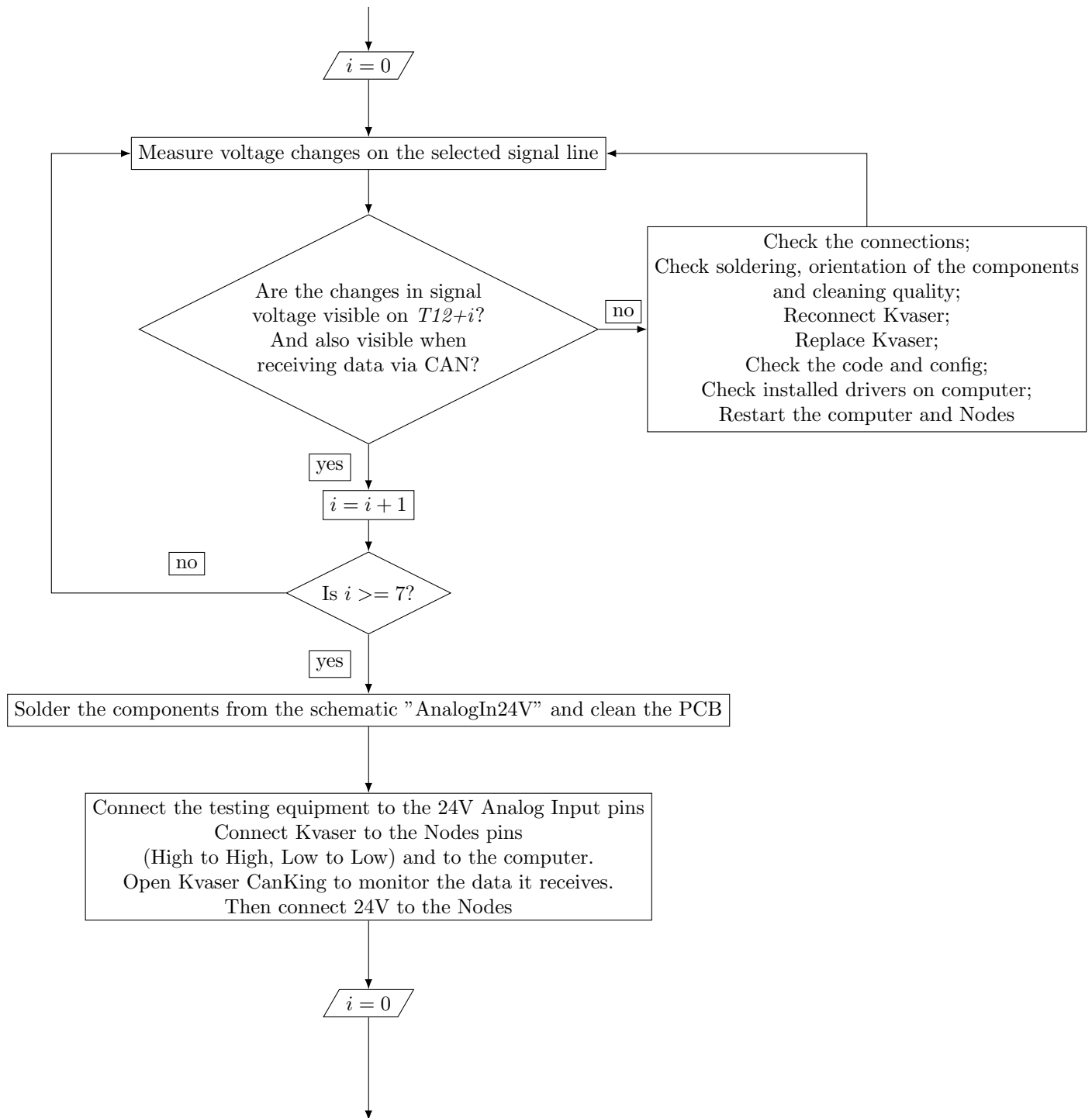


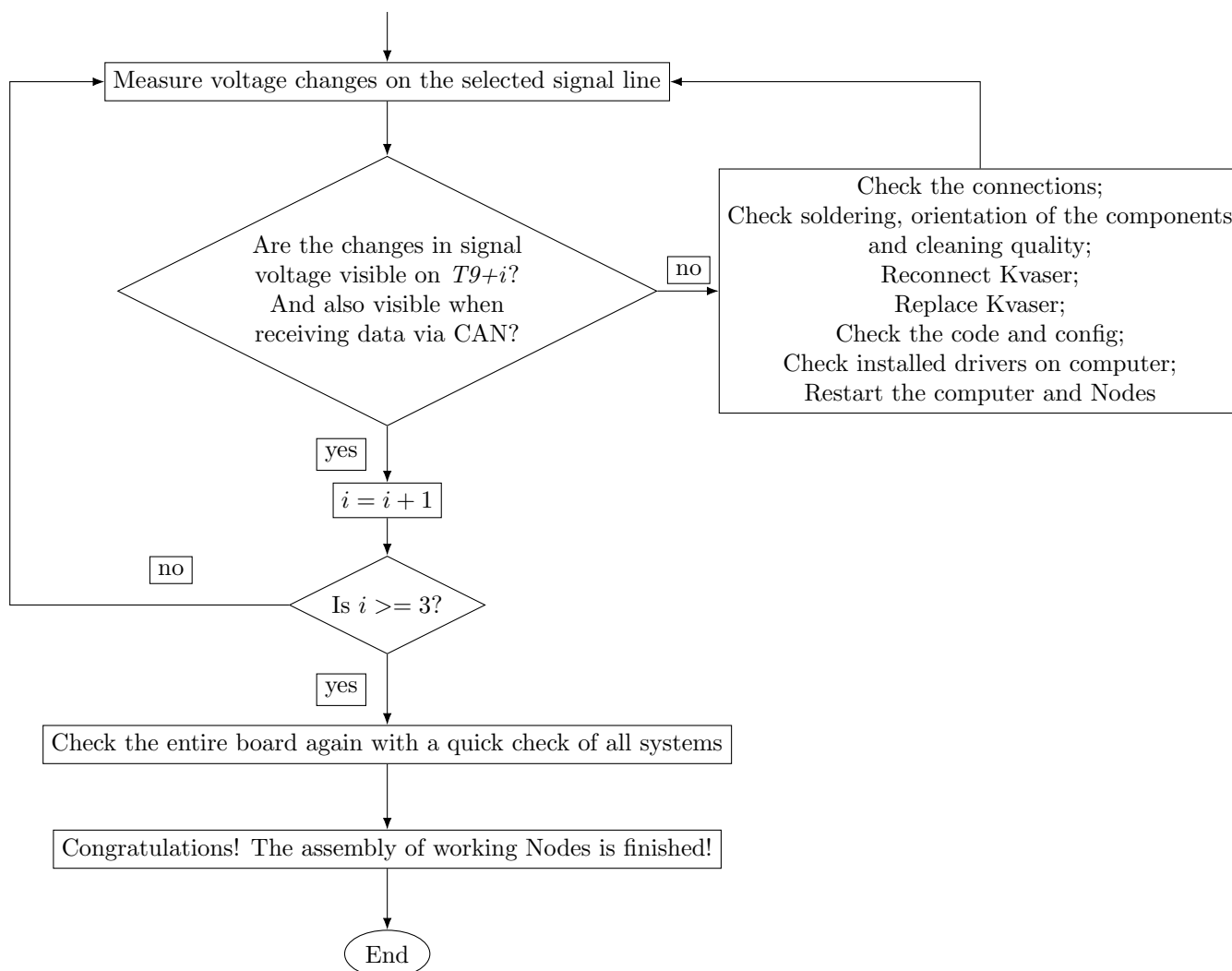












Package Description

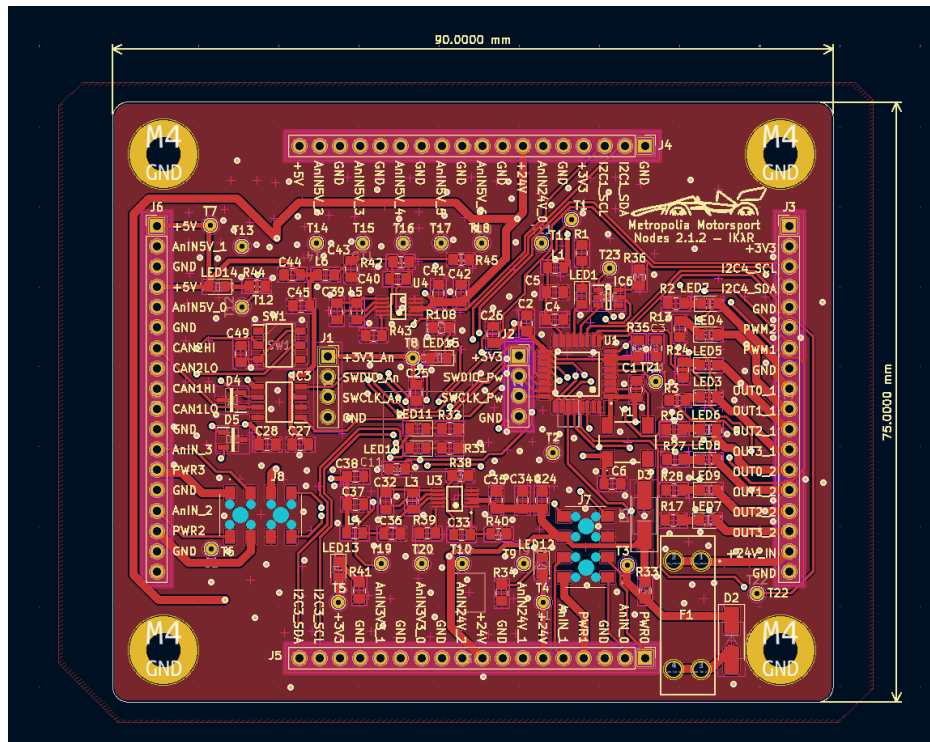


Figure 26: Dimensions Nodes v2.1

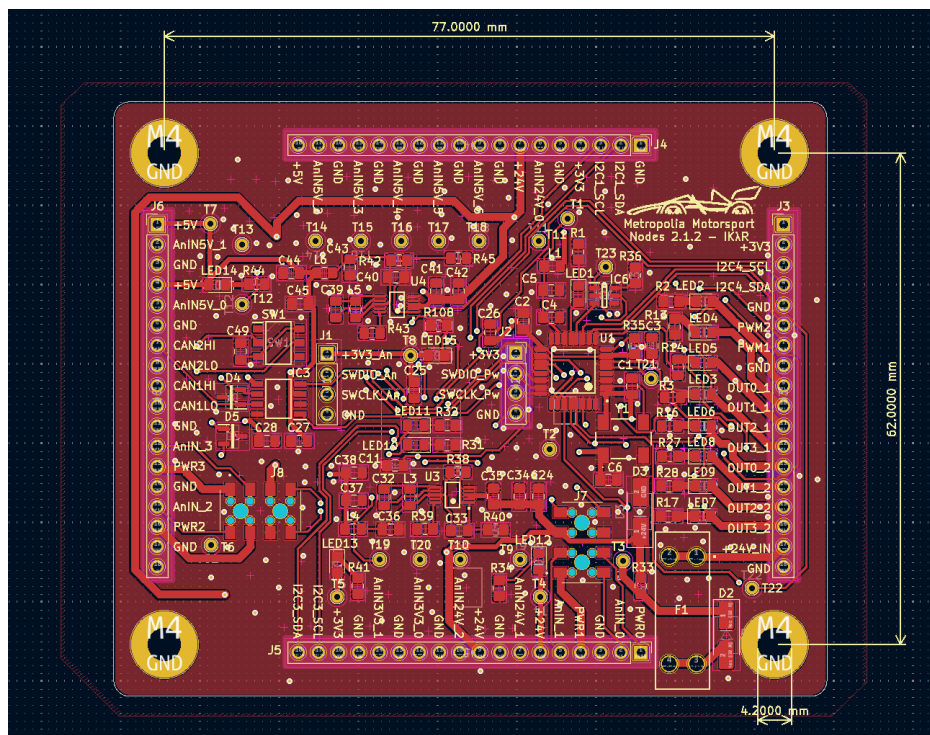


Figure 27: Mounting dimensions Nodes v2.1

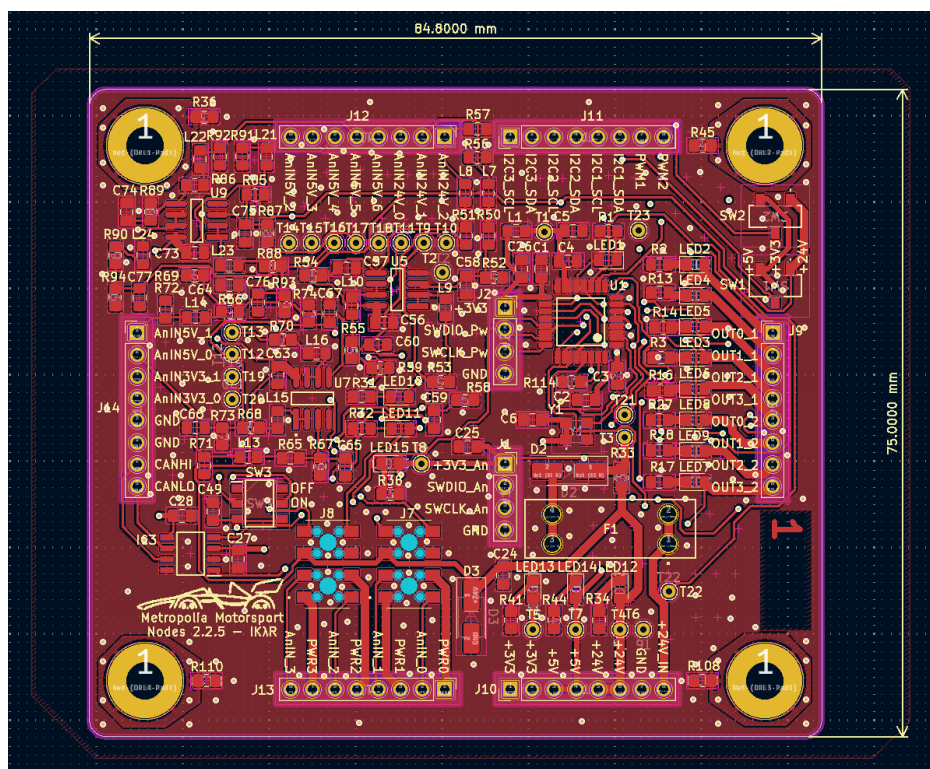


Figure 28: Dimensions Nodes v2.2

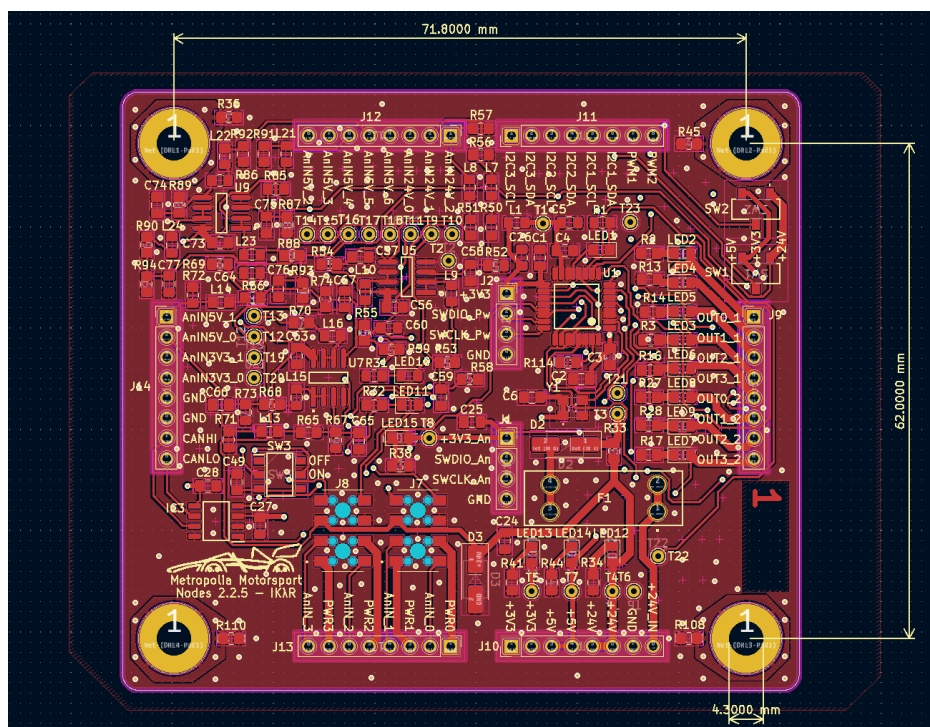


Figure 29: Mounting dimensions Nodes v2.2

Bill Of Materials

This BOM corresponds to the latest version of Nodes.

Table 12: Bill Of Materials

Ref.	Name	Sold. Temp.	Value	Footprint	Docs.	Qty.	Descr.
C1, C2, C6, C7, C8, C9, C10, C12, C13, C14, C15, C17, C19, C22, C25, C26, C27, C28, C29, C32, C47, C48, C56, C63, C68, C73, C78, C81	08055C104KAT4A	$260^{\circ}C \pm 5^{\circ}C$ $10 \pm 0.5 \text{ sec}$	100nF	0805	[20]	28	
C4, C20, C64, C65, C66, C67, C69, C70, C71, C72, C74, C75, C76, C77, C79, C80, C82, C83, C84, C85	C0805C103KARACAUTO	$260^{\circ}C$	10nF	0805	[21]	20	
C57, C58, C59, C60, C61, C62	CL21B102KBCNNNC	$245 \pm 5^{\circ}C$ $3 \pm 0.3 \text{ sec}$	1nF	0805	[22]	6	
C3, C11, C24	GRM21BZ71H475KE15L	$270 \pm 5^{\circ}C$ $10 \pm 0.5 \text{ sec}$	4.7uF	0805	[23]	3	
C5, C16, C21	08053C105KAZ2A		1uF	0805	[24]	3	
C30, C31	CL21B474KBFNNNG	$245 \pm 5^{\circ}C$ $3 \pm 0.3 \text{ sec}$	470nF	0805	[22]	2	
C18	CL21B225KBYNNWE		2.2uF	0805	[26]	1	
C23	CL21B106KAYQNNE	$245 \pm 5^{\circ}C$ $3 \pm 0.3 \text{ sec}$	10uF	0805	[22]	1	
C49	AC0805KRX7RBBB472	$260 \pm 5^{\circ}C$ $10 \pm 0.5 \text{ sec}$	4.7nF	0805	[25]	1	
R4, R5, R7,	RT0805CRE0715KL	$260^{\circ}C$	15k Ω	0805	[27]	16	

References	Name	Sold. Temp.	Value	Footprint	Docs.	Qty.	Descr.
R8, R9, R10, R11, R12, R18, R19, R21, R22, R23, R24, R25, R26		10 sec					
R44, R50, R51, R60, R65, R66, R75, R76, R85, R86, R95, R100, R101, R102, R105	ERJ-PB6B1001V	270°C 10 sec	1kΩ	0805	[28]	15	
R56, R57, R63, R71, R72, R81, R82, R91, R92, R97, R106, R107	CRCW080510M0FKEA	260 ± 5°C 10 ± 1 sec	10MΩ	0805	[29]	12	
R58, R59, R64, R73, R74, R83, R84, R93, R94, R99, R111, R117	ERJ-PB6B3241V	270°C 10 sec	3.24kΩ	0805	[28]	12	
R2, R3, R13, R14, R16, R17, R27, R28, R34	CR0805-FX-5621ELF	270 ± 5°C 10 ± 1 sec	5.62kΩ	0805	[31]	9	
R68, R70, R78, R80, R88, R90, R96	ERJ-PB6B1741V	270°C 10 sec	1.74kΩ	0805	[28]	7	
R112, R113, R114, R115, R116, R118	AC0805FR-7W10KL	260 ± 5°C 10 ± 1 sec	10kΩ	0805	[30]	6	
R1, R31, R32, R38, R41	ESR10EZPJ301	260 ± 5°C 10 ± 1 sec	300Ω	0805	[32]	5	
R6, R15, R20, R29	CR0805-FX-2201ELF	270 ± 5°C 10 ± 1 sec	2.2kΩ	0805	[31]	4	
R36, R45, R108, R110			0Ω	0805		4	
R37, R46, R47, R109	RC0805FR-13120RL	260 ± 5°C 10 ± 1 sec	120Ω	0805	[33]	4	
R53, R55, R62	ERJ-PB6B2102V	270°C	21kΩ	0805	[28]	3	

References	Name	Sold. Temp.	Value	Footprint	Docs.	Qty.	Descr.
		10 sec					
R33, R35	WSL0805R0100FEA18	260°C 10 sec	10mΩ	0805	[34]	2	
R39, R42	TNPW08051M00BEEN	260 ± 5°C 10 ± 1 sec	1MΩ	0805	[35]	2	
R40, R43	ERJ-6ENF5763V	270°C 10 sec	576kΩ	0805	[36]	2	
R30	RT0805DRE074K7L	260°C 10 sec	4.7kΩ	0805	[27]	1	
R48	ERA-6AEB8872V	270°C 10 sec	88.7kΩ	0805	[37]	1	
R49	ERA-6AEB2212V	270°C 10 sec	22.1kΩ	0805	[37]	1	
L7, L8, L9, L10, L11, L12, L13, L14, L15, L16, L17, L18, L19, L20, L21, L22, L23, L24, L25, L26, L27, L28, L29, L30	LQM21PN1R0NGCD		1uH	0805	[38]	24	
L1, L2	BDCC00201208R33MWA		330nH	0805	[39]	2	
L5	CV201210-4R7K	260°C 10 sec	4.7uH	0805	[40]	1	
D5, D6	1SMA5913BT3G	260°C 10 sec		SMA	[1]	2	
D1	S1M-13-F			SMA	[41]	1	
D2	SX33L_R1.00001			SMA	[3]	1	
D3	1SMA5934BT3G	260°C 10 sec		SMA	[1]	1	
D4	ESDCAN24-2BLY	240 – 245°C 60 sec		SOT-23-3	[41]	1	
U5, U6, U7, U8, U9, U10	LM358BAIDR			SOIC-8	[17]	6	

References	Name	Sold. Temp.	Value	Footprint	Docs.	Qty.	Descr.
U1	STM32G431KBT6			LQFP-32	[8]	1	
U11	STM32G431RBT6			LQFP-64	[8]	1	
Y1, Y2	SG-8018CE_8.0000M-TJHPA3				[10]	2	
F1	3557-2				[43]	1	
SW1, SW2	CJS-1201TB1				[44]	2	
SW3	CAS-220TB1				[45]	2	
LED2, LED5, LED6, LED9, LED12	SML-210VTT86			0805	[46]	5	Red LED
LED3, LED4, LED7, LED8	SMLMN2WB1CW1	325°C 3 sec		0805	[47]	4	White LED
LED1, LED13	SML-210YTT86M			0805	[46]	2	Yellow LED
LED10, LED14	SML-H12P8TT86	400°C 3 sec		0805	[48]	2	Green LED
LED11	SML-H12D8TT86	400°C 3 sec		0805	[49]	1	Orange LED
LED15	150080BS75000	260°C 10 sec		0805	[50]	1	Blue LED
IC1, IC2	VNQ9025AJTR			Power SSO-16	[16]	2	
IC3, IC4	TCAN1462VDRQ1			SOIC-8	[19]	2	
IC5, IC6	ST730MR			SOT-23-5	[5]	2	
Q1, Q2	TN2106K1-G			SOT-23-3	[13]	2	
PS1	LMR50410YFQDBVRQ1			SOT-23-6	[6]	1	
J9, J10, J11, J12, J13, J14						6	1x8
J1, J2						2	1x4
J7, J8	HLE-104-02-L-DV-BE-A-P-TR				[51]	2	2x4

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